



Designing a perishable closed-loop poultry supply chain: metaheuristic approaches and model evaluation

Zahra Akbari-Aghghaleh¹ · Ashkan Mozdgir¹ · Iman Seyedi² · Enza Messina²

Received: 16 July 2024 / Accepted: 20 July 2025

© The Author(s), under exclusive licence to Springer Nature B.V. 2025

Abstract

In this study, a cutting-edge approach focusing on pivotal innovation to design a closed-loop poultry supply chain network is developed. As the poultry sector faces complex challenges regarding cost management and customer service enhancement, we propose a novel mixed-integer linear programming framework specifically designed for designing closed-loop poultry supply chains. The proposed framework introduces flexibility in the case of a closed-loop poultry supply chain, while most of the perishable products outside the poultry industry, like red meat, dairy products, seafood, and fresh produce, can be helped through visibility and optimization strategies that can help cross-industrial boundaries. In this regard, the objective is to minimize the supply chain cost. The model is widely evaluated using simulated data with twenty-four test problems of varying scales. Recognizing the NP-hard characteristics of the developed model, which make it impractical for large-scale problems, this limitation is addressed by incorporating four established metaheuristic algorithms, namely Differential Evolution, Simulated Annealing, and Genetic Algorithm. Additionally, we explore two hybrid combinations of these algorithms, namely Hybrid Genetic Algorithm with Simulated Annealing and Hybrid Differential Evolution and Simulated Annealing Algorithm. The results of comprehensive test problems and sensitivity analysis bring out the remarkable efficacy of our proposed network design model, pointing out its prospects for optimizing closed-loop poultry supply chain network operations. Specifically, Hybrid Genetic with Simulated Annealing Algorithm in small cases, and Differential Evolution in medium and large-size cases emerge as the best-performing algorithms that perform superiorly in optimizing supply chain networks across different problem sizes.

Keywords Poultry supply chain · Closed-loop supply chain network design · Perishable product · Hybrid metaheuristic algorithms · Mixed-integer linear programming

Extended author information available on the last page of the article

1 Introduction

The importance of perishable products in traditional industries, including fresh, agricultural, and livestock products, extends to pharmaceutical items. According to Amorim et al. (2013), the main feature that characterizes all the valuable perishable products is the fast decrease in market value while their consumption value decreases slowly. This evidence shows that items of such kinds, when unsold, retain high salvage values, which could be further developed and reused during successive periods. Terms of the closed-loop supply chain (CLSC) lifecycle of the CLSC mainly revolve around end-of-life product management; they comprise direct reuse, recycling, remanufacturing, and centralized waste disposal. Remanufactured products can reduce costs by up to 50% compared to new products. Also, this can reduce energy, materials, and pollutant emissions by 60%, 70%, and 80%, respectively (Ostojic, 2016).

Numerous operations researchers have focused on improving individual layers or the entire supply chain (SC) by creating novel models and methods in recent years. A pivotal focus for companies striving to attain goals related to time efficiency, economic competitiveness, and service quality is Supply Chain Management (SCM) (Seyedi et al., 2016). Designing an SC is one of the most important considerations when making short-term, medium-term, and long-term decisions within the SC. The aim of this decision-making process is to optimally allocate and position resources and facilities. A framework for assessing sustainability has been created to enhance transparency within global supply chains (Okay et al., 2024). Environmental factors have also become a significant focus within the supply chain. Practitioners in the supply chain expect academic research to explore innovative extensions and effective problem-solving methods (Govindan et al., 2015). In 2018, the agricultural sector emitted 9.3 billion tons of CO₂ worldwide (FAO, 2020). Beyond challenges like unemployment, consumer health and safety, food security, and poverty, the operations of food supply chains have a notable impact on community development. The livestock sector is a major contributor to greenhouse gas emissions and other consequences. Poultry, which encompasses meat and eggs, is the fastest-growing segment of the livestock industry globally (Li et al., 2022).

As observable in Fig. 1, poultry production has encompassed a more significant share than other animal products. According to De Vries et al. (2013), the human population is projected to surpass 9 billion in 2050. The reason is that with food demand and consumption on the rise, the livestock sector becomes the primary source of greenhouse gas emissions and other environmental impacts. Poultry production, whether for meat or eggs, constitutes one of the rapidly growing food production sections (Conway, 2015).

Taking this into account, logistics establishes CLSCs by taking into consideration both the forward and reverse flows at the same time (Pishvaei et al., 2011). The design of CLSCs is a vital issue that involves quantity, placement, capability, cooperation of facilities, products, value addition, and inventory control decisions for integrating all operations (Baghalian et al., 2013). In the last decade, financial profits and the environmental effects of waste materials have persuaded many companies to focus on CLSCs (Mavi et al., 2023; Seyedi et al., 2012). Initially, waste management and environmental issues were not SC management's concern; however, recently, RL has turned out to be a fascinating topic. A CLSC has emerged, considering both forward and reverse SCs (Govindan & Soleimani, 2017). SCs have recently been subject to many changes, and firms have recently been facing new chal-

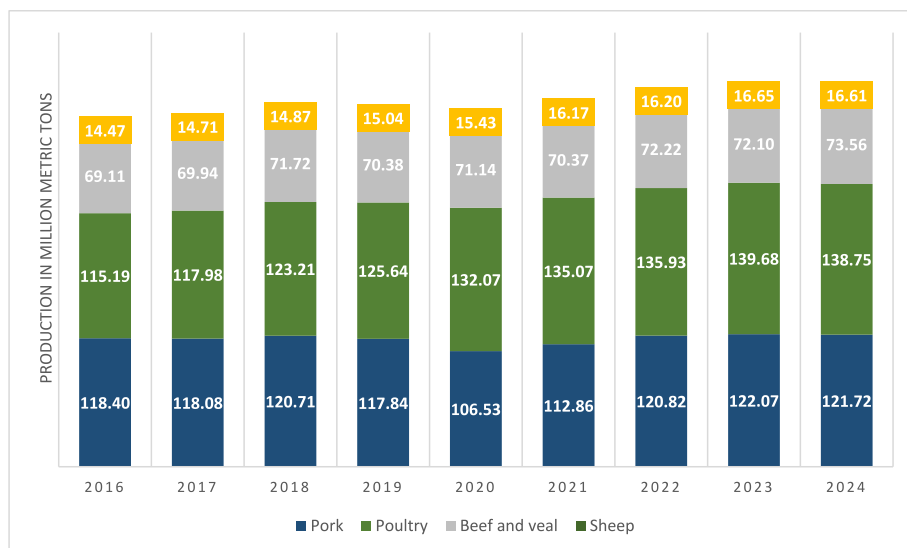


Fig. 1 Poultry production compared to other meats from 2016 to 2024 (in million tons). Source: <https://www.statista.com>

lenges while trying to satisfy consumers to be competitive in the global markets (Shekarabi et al., 2024).

As illustrated by both Fig. 1 and Fig. 2, on the global platform, there is a contracting upwards in demand for poultry production to 2024. It, therefore, becomes of absolute importance, with such a significant volume of exportable products involved, that those factors influencing the export of such products be considered. Among these factors, poultry products' health and environmental impacts are the most prominent issues affecting poultry product exports. Produced chicken encompasses meat, skin, blood, bones, feathers, poultry

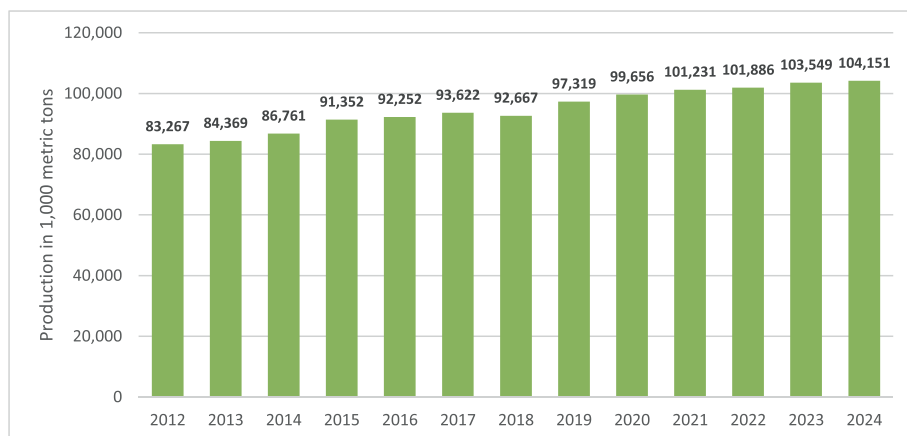


Fig. 2 Global chicken meat production 2012–2024. Source: <https://www.statista.com>

bedding materials, and more throughout the production, rearing, processing, and distribution processes. These by-products constitute a significant percentage of the weight of the produced chicken; hence, a considerable amount of these waste materials becomes evident annually. Incineration and natural decomposition primarily serve as methods for disposing of these wastes. Poultry waste does not contain heavy metals or disease-causing agents; however, it may significantly affect the surrounding soil and air quality. Table 1 demonstrates the poultry production per capita for various countries worldwide.

This research focuses on designing a CLSC for poultry logistics networks, integrating producers, distribution centers, processing facilities (e.g., processed poultry centers), and post-processing facilities (e.g., poultry powder production). The final products are delivered to customers, including fresh poultry markets, processed poultry markets, and poultry powder markets.

The model assumes fixed locations for customers and processing centers. The reverse flow involves returned products, which are shipped to post-processing centers to be converted into by-products. These by-products are then redirected to recycled product streams or poultry powder markets. Notably, livestock farms—potential consumers of poultry powder—may also act as customers in the CLSC network.

By applying advanced optimization techniques to production and distribution phases, our model aims to enhance efficiency across all supply chain tiers while maintaining low costs and competitiveness. This approach equips local farmers with a more viable and resilient business model, enabling them to withstand low-cost import pressures and maintain operations in challenging market conditions.

Beyond the poultry industry, the proposed CLPSCN framework is adaptable to other perishable and time-sensitive products, such as red meat, seafood, and fresh products. The model addresses critical challenges common to these sectors, including the NP-hard complexity of large-scale supply chain optimization and perishability constraints. To solve these challenges, the model integrates advanced meta-heuristic algorithms—Differential Evolution (DE), Simulated Annealing (SA), and Genetic Algorithm (GA)—alongside hybrid variants. These methods enable efficient exploration and exploitation of the solution space, ensuring near-optimal solutions within a reasonable computational time. Rigorous testing across diverse problem sizes demonstrated the algorithms' superior performance, particularly in minimizing total supply chain costs while maintaining operational efficiency and responsiveness.

Table 1 Poultry production per capita by world countries in 2023 (source: <https://www.statista.com>)

Rank	Country	Number of produced chicks
1	China - Mainland	110,901,585,000
2	United States of America	9,325,584,000
3	Brazil	6,184,124,000
4	Indonesia	4,573,210,000
5	India	2,732,466,000
6	Russian Federation	2,448,906,000
7	Mexico	2,013,347,000
8	Iran	1,881,311,000
9	Greece	1,609,832,000
10	Pakistan	1,533,000,000

Given these results, the model proves highly applicable not only to poultry logistics but also to broader perishable goods supply chains, where cost management and service-level optimization are pivotal.

2 Literature review and research background

The review provides a comprehensive study of the PSC along with the applications of CLSC and RL. SCM concepts are widely applicable across various industries, such as education, mining, automotive, electricity, agriculture, horticulture, healthcare, pharmaceuticals, and more (Hasani et al., 2012). Designing SC networks is challenging, as various environmental, economic, social, and political factors are considered, along with many other factors, to gain better benefits. Logistics would be crucial in optimizing all inventory materials with transportation and information management, increasing the SC's success rate for the developed and developing industries.

Recent studies emphasize that inappropriate policies, wrong information, and lack of awareness can also cause global warming and environmental problems, such as impure air, water salinity, and loss of top texture in the soil. Thus, nowadays, researchers have developed a keen interest in RL and CLSC. The available past relevant research helps us understand the dimensions of CLSC more constructively. Literature reviews show that about 80% of the published papers propose mathematical models for studying RL or CLSC while highlighting the gaps comprehensively. Shekarian (2020) classified different articles using game-based studies based on influential factors in CLSC models. These studies considered quality, inventory, economic, environmental, financial, and commercial factors. The review, categorization, and assessment of all relevant 83 impacts were done by Govindan & Soleimani (2017). Their work highlighted the past and may develop further research areas in CLSC.

In recent years, hundreds of researchers have analyzed both perishable and non-perishable products/SCs, focusing their attention mainly on ripe fruits (Cheraghalipour et al., 2018), dairy products (Sel et al., 2015), and agriculture (Borodin et al., 2016).

Rajabi-Kafshgar et al. (2023) suggested a hybrid metaheuristic to solve a closed-loop agricultural SC network design problem. Pistachios are selected as a case study in this regard. They found the solution to the mathematical model by using metaheuristic algorithms, where minimum total cost and social issues, along with maximum corporate social responsibility, were considered the objective functions. This article elaborates on how wastes are turned into compost and oil, which goes back to the factory. Tavana et al. (2022) address a multi-objective MILP model for the design of sustainable CLSC with interconnections. Its performance is tested through simulated data in small to medium-sized test problems. The results related to test problems and sensitivity analyses prove and validate the effectiveness of the proposed model. Baghizadeh et al. (2021) proposed a mixed-integer nonlinear bi-objective optimization model to minimize network costs and maximize social scales and flexibility. The problem has been taken in a closed-loop manner for the reproduction case study, which comprises four levels: the optimal location of distribution centers, collection points, recycling locations, and fixed factory points.

Hasani et al. (2012) presented extensive work on developing a robust CLSC for perishable products. They present a mixed-integer linear programming model, and their sug-

gested framework considers a range of assumptions like multiple periods, multiple types of products, multiple supply systems, random demand, and purchasing costs. Yavari & Geraeli (2019) proposed an exploratory method for a powerful optimization model in designing a green CLSC related to dairy products. This multi-period and multi-product model includes supplier levels, producers and warehousing, retailers, and collection centers. They use MILP and an innovative MILP robust model while developing an efficient heuristic method called YAG for solving large-scale problems. Computational experiments show the efficiency of the performed approach in cost minimization and environmental pollutants. Compared with the deterministic models, the robust model demonstrated its ability to stand against any disruption.

Rohmer et al. (2019) examined the design of a sustainable supply chain network in the nutrition-sensitive food system by taking a multi-objective approach toward meat production, processing, and agricultural transportation. They presented the model through the epsilon-constraint method and compromise programming, which brought satisfactory results from the objectives of the problem: minimization objectives that concern economic and environmental aspects. Jouzdani & Govindan (2021) presented a multi-objective mathematical programming model to minimize costs, energy consumption, and traffic density related to SC operations. Among other issues, they detected and analyzed the interdependencies and interactions of three sustainability aspects known as the triple bottom line, considering several vehicles and various products. In this current study, product lifetime uncertainty was explicitly modeled as a Weibull random variable; their research proposed a chance-constrained mathematical programming model for designing a multi-period sustainable perishable food SC in the case study of packaged dairy products.

Mosallanezhad et al. (2021) investigate and propose the design of a CLSC for shrimp and subsequently offer a mathematical model aimed at minimizing the total cost across the SC. Due to its inability to handle large-sized problems, the model employs three well-known heuristics and two hybrid combinations. Additionally, fifteen test cases are generated to validate the model in a real-world application. Salehi-Amiri et al. (2022) present a model for designing a closed-loop avocado SC network. The case study in Brazil examines the proposed MILP model's effectiveness in determining optimal locations for new facilities, avocado flow to customers and various centers, and optimal employment in the opened facilities.

Khalilpourazari & Arshadi Khamseh (2019) explored and proposed a new multi-objective mathematical model for an effective blood SC design in earthquakes. This paper employs two transportation modes with different speeds and capacities for blood transfer from collection centers to blood centers while addressing constraints on available conveyors. This study further illustrates the efficacy of the proposed model in designing cost and time-efficient blood supply chains using various multi-objective decision-making methods in practical applications during severe earthquakes. Critical parameters, which have significant effects on objective functions, are also brought out using sensitivity analyses. Seyfi-Shishavan et al. (2021) suggested a methodology that could minimize the sum of the total cost of the blood supply network and total blood shortage. They have proposed a fuzzy optimization model for designing a resilient blood supply chain network, focusing mainly on handling blood distribution in the case of earthquakes and similar other scenarios. This approach pursues minimizing total costs and blood shortage by proposing a novel fuzzy multi-period mathematical model using trapezoidal fuzzy numbers, whose uncertain parameters are rep-

resented by spherical fuzzy sets. The proposed model is oriented towards a holistic approach to dealing with uncertainties in managing blood supply chains, considering the possibility of natural disasters in Istanbul, and incorporating the concept of disruption throughout the supply chain. Numerous studies have made significant contributions to the management of uncertainty in complex systems. Özmen et al. (2017) enhanced the robustness of regulatory models by addressing polyhedral uncertainty, thereby improving reliability in the face of parametric variations. Özmen et al. (2014) developed spline regression models utilizing MARS and CMARS to tackle parameter fluctuations in multi-modal systems. Researchers such as Weber et al. (2008) and Kuter et al. (2018) have employed mathematical and statistical techniques to estimate parameters under uncertain conditions. Additionally, machine learning approaches have gained traction for parameter estimation in uncertain environments, as demonstrated by Çevik et al. (2017), Khalilpourazari et al. (2020), Tashakkori et al. (2024), Haghbayan et al. (2023), Rajabi-Kafshgar et al. (2025), and Karimi et al. (2024). These methodologies offer robust solutions for managing uncertainty, particularly in the context of CLSC optimization.

Hamdan & Diabat (2020) proposed a bi-objective mathematical model to design a disaster-resilient blood SC. The intention was to present a two-stage stochastic optimization model to minimize the time and cost of blood delivery to hospitals in case of a disaster. For this purpose, potential failures have been considered in all the blood facilities and transportation routes. The model is then solved using the Lagrangian relaxation algorithm; it is applied to a real case study of blood banks in Jordan. Kaya & Ozkok (2020) presented an efficient SC network for blood distribution to meet regional hospital demands. They presented a MINLP model in which the decisions are for determining the optimal selection, Blood Bank, the hospital allocation to Blood Banks (LBBs), optimal inventory level in each LBB, and routing decisions among facilities that minimize the overall system cost. It is suggested that this problem be solved using a linear piecewise approximation method and an exploratory method like simulated annealing. It then applies it to the blood distribution network in Istanbul and shows significant improvements over the existing design.

Fallahi et al. (2021) first developed a CLSC for blood products, taking into consideration the transportation equipment for blood and their respective quality attributes, after which they solved the problem by using a DE algorithm, including two new variants of DE. Finally, they draw some managerial implications through sensitivity analyses of the model's parameters. Reduction in procurement costs, increased resilience, and decreased disposal costs are practical advantages of Closed-Loop Reverse Logistics (RTL). Considering the complexity of the blood SC problem, an evolutionary algorithm, Differential Evolution, is suggested for solving the proposed Closed-Loop Blood SC problem. Karadağ et al. (2021) delve into redesigning a mobile unit-enabled blood SC organization. Experts prioritize the project goals using hierarchical priority analysis. The final comparison indicates that the proposed model exhibits a more effective solution by 0.25%. Fallahi et al. (2022) focus on examining and designing an efficient SC for managing convalescent plasma during the COVID-19 pandemic. Their study aims to plan an integrated green SC network for regular and convalescent plasma during the pandemic, minimizing total costs and carbon emissions. The model is applied to a real case study in Iran. The research presents a four-tier, multi-period, multi-product, multi-objective model for designing a plasma SC during the COVID-19 pandemic. Yang & Chen. (2021) presented a value-driven CLSC in an uncertain environment, focusing on perishable products. The SC involves four tiers: producers,

wholesalers, recycling centers, and consumer markets. Optimal solutions are obtained in harsh and favorable environments with a numerical example, indicating the proposed method's superior precision and more robust selection in location compared to nonlinear integer programming. Shokouhifar et al. (2024) propose a new method by developing an exergetic mathematical model comprising economic, environmental, and social dimensions within a six-echelon closed-loop food supply chain. The paper has also proven that EEA and SA can reduce total exergy consumption by combining WOA and metaheuristic ensemble algorithms. Applying this model in a real-world dairy supply chain in Iran will enable more practical merits of this integration approach, using exergy analysis and state-of-the-art optimization techniques to enhance sustainability across supply chains. Table 2 provides a summary of related articles.

2.1 Research gap

Based on the previous section, a thorough investigation encompassing SC, RL, and CLSC has been considered. The implementation of CLSCs on perishable and non-perishable products and the benefits they offer to numerous businesses have led many investors to apply simple or complex network designs based on this concept. This section shows that multiple studies aim to determine an optimal point for their network costs in a CLSC. Commonly, these networks are approved to address practical concerns such as satisfying customer demands, transporting various materials to target markets, determining the most efficient and cost-effective routes, optimizing inventories, and identifying the optimal location for opening new facilities. Specifically, the use of RL and CLSC in the livestock product industry, especially poultry, is evident but often overlooked.

Another noteworthy aspect highlighted in the reviewed literature is that designing the CLSC for the poultry business has not been previously investigated. The primary advantages of developing such networks include minimizing the cost of the chain. This research differs from previous works in that many prior studies focused on creating comprehensive or realistic networks for other products, while this research fills the gap by addressing perishable products. Since developing a new SC for each product, significant headaches such as delivery, overall cost management, and procurement have been cushioned to a certain extent.

Considering a broader perspective, most prior researchers aimed to fill the gap in SC material waste by using old logistics concepts. Applying SC characteristics in the perishable goods industry may be more effective than conventional techniques. Therefore, numerous studies have implemented SC networks for perishable goods to minimize transportation expenses and logistical concerns and optimize various product values. While potential costs are reduced and additional income is generated for their users, many perishable products can be recycled back into the chain to extract added value. Accordingly, considering CLSC for perishable products like poultry benefits both forward-looking profits and reverse-added value.

3 Modeling and solution approach

In this section, a mathematical model of poultry CLSC is developed. The solution approach of the developed poultry CLSC is also described.

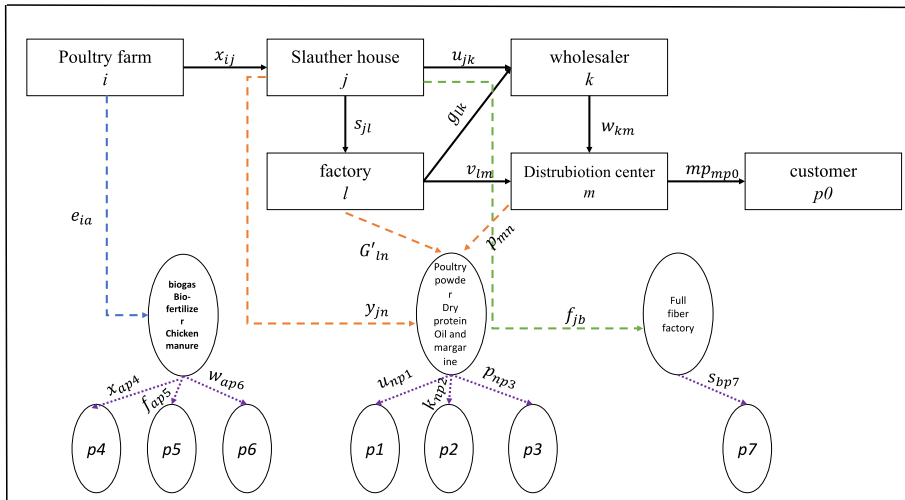


Fig. 3 Layers of the closed-loop PSC

3.1 Mathematical model

The current study aims to design a single-cycle PSC network comprising producers, slaughterhouses, wholesalers, processing centers (factories), distribution centers, recycling plants, poultry and livestock food markets, and customer markets. As illustrated in Fig. 3, raised poultry are transported to slaughterhouses. Depending on their capacity, slaughterhouses send poultry to wholesalers and processing factories. Furthermore, after skinning, processing, packaging, and freezing, these factories must transport the final goods to distribution centers and wholesalers. Eventually, poultry waste collected at each stage from farms, slaughterhouses, processing factories, and distribution centers is transported to waste recycling plants. The recycled products are sent to customer markets for recycled product demand. Since poultry farms (producers) are considered potential customers for livestock and poultry feed, the network could be regarded as a CLSC, with producers acting as customers for poultry meal. The Schematic view of the closed-loop PSC layers is visible in Fig. 3.

The indices, parameters, and decision variables for the mathematical model are as follows:

Indices

Location of Poultry Production (Poultry Farms)	$i = 1, 2, \dots, I$
Potential Slaughterhouse Location	$j = 1, 2, \dots, J$
Potential Wholesale Center Location	$k = 1, 2, \dots, K$
Potential Poultry Processing Factory Location	$l = 1, 2, \dots, L$
Potential Distribution Center Location (Identified and Fixed)	$m = 1, 2, \dots, M$
Potential Poultry Waste Recycling Plant Location	$n = 1, 2, \dots, N$
Potential Biomass Recycling Plant and Organic Fertilizer and Compost Factory Location	$a = 1, 2, \dots, A$
Potential Feather Processing Factory Location	$b = 1, 2, \dots, B$
Primary Non-Waste Poultry Products (including processed products in Factory L and products produced in the slaughterhouse)	p
Products Produced from Poultry Waste: Poultry Meal	s_1

Products Produced from Poultry Waste: Dry Protein	s_2
Products Produced from Poultry Waste: Oil and Margarine	s_3
Products Produced from Poultry Waste: Biomass	o_1
Products Produced from Poultry Waste: Organic Fertilizer	o_2
Products Produced from Poultry Waste: Compost	o_3
The product produced from poultry waste: Feather fibers	t
Markets for non-waste poultry products	$p_0 = 1, 2, \dots, P_0$
Markets for poultry meal	$p_1 = 1, 2, \dots, P_1$
Markets for dry protein	$p_2 = 1, 2, \dots, P_2$
Markets for oil and margarine	$p_3 = 1, 2, \dots, P_3$
Markets for biomass	$p_4 = 1, 2, \dots, P_4$
Markets for organic fertilizer	$p_5 = 1, 2, \dots, P_5$
Markets for compost centers	$p_6 = 1, 2, \dots, P_6$
Markets for feathers	$p_7 = 1, 2, \dots, P_7$
Parameters	
Fixed cost of opening a poultry farm and chicken house, denoted by	fc_i
The fixed cost of opening a poultry slaughterhouse center is denoted by j	fk_j
The fixed cost of opening a poultry processing factory is denoted by l	fb_l
The fixed cost of opening a wholesale center is denoted by k	fm_k
The fixed cost of opening a poultry waste recycling factory is denoted by n	fj_n
The fixed cost of opening a biomass and fertilizer recycling factory is denoted by a	fl_a
The fixed cost of opening a feather recycling factory is denoted by b	fv_b
Transportation Costs per Unit of Product:	
Transportation cost per unit from poultry producer i to slaughterhouse j	cx_{ij}
Transportation cost per unit from slaughterhouse j to wholesale center k	cu_{jk}
Transportation cost per unit from slaughterhouse j to processing factory l	ca_{jl}
Transportation cost per unit from wholesale center k to distribution center m	cb_{km}
Transportation cost per unit from processing factory l to wholesale center k	cs_{lk}
Transportation cost per unit from processing factory l to distribution center m	col_{lm}
Transportation cost per unit from distribution center m to market p_0	cp_{0mp0}
Transportation cost per unit of poultry waste from slaughterhouse j to recycling factory n	cp_{jn}
Transportation cost per unit of poultry waste from processing factory l to recycling factory n	cd_{ln}
Cost of transportation per unit of poultry waste from distributor m to recycling plant n	ck_{mn}
Cost of transportation per unit of poultry waste from farm i to biogas and fertilizer production center a	cj_{ia}
Cost of transportation per unit of slaughterhouse waste from slaughterhouse j to reprocessing plant b	cm_{jb}
Cost of transportation per unit of product s1 from recycling plant n to customer market p1	cls_{1np1}
Cost of transportation per unit of product s2 from recycling plant n to customer market p2	cls_{2np2}
Cost of transportation per unit of product s3 from recycling plant n to customer market p3	cls_{3np3}
Cost of transportation per unit of product o1 from recycling plant a to customer market p4	clo_{1ap4}
Cost of transportation per unit of product o2 from recycling plant a to customer market p5	clo_{2ap5}
Cost of transportation per unit of product o3 from recycling plant a to customer market p6	clo_{3ap6}

Cost of transportation per unit of processed product t from reprocessing plant b to customer market p7	clt_{bp7}
Cost of producing per unit of product at poultry producer center i	cp_i
Cost of processing per unit of product at slaughterhouse center j	cpj_j
Cost of processing per unit of product at processing center l	cpl_l
Cost of processing per unit of product at wholesale center k	cpk_k
Cost of producing per unit of product at distribution center m	cpm_m
Cost of producing per unit of product s1 at recycling plant n	$cpns_{1n}$
Cost of producing per unit of product s2 at recycling plant n	$cpns_{2n}$
Cost of producing per unit of product s3 at recycling plant n	$cpns_{3n}$
Cost of producing per unit of product o1 at recycling plant a	$cpao_{1a}$
Cost of producing per unit of product o2 at recycling plant a	$cpao_{2a}$
Cost of producing per unit of product o3 at recycling plant a	$cpao_{3a}$
Cost of producing per unit of processed product at recycling plant b	cpb_b
Production capacity of poultry breeder i	λI_i
Production capacity at slaughterhouse center j	λJ_j
Production capacity at processing factory l	λL_l
Storage capacity at processing factory l	$\lambda L'_l$
Storage capacity at slaughterhouse center j	$\lambda J'_j$
Storage capacity at wholesale center k	λK_k
Storage capacity at distribution center m	λM_m
Production capacity at recycling plant n for product s1	λn_{1ns1}
Production capacity at recycling plant n for product s2	λn_{2ns2}
Production capacity at recycling plant n for product s3	λn_{3ns3}
Production capacity at recycling plant a for product o1	λa_{1ao1}
Production capacity at recycling plant a for product o2	λa_{2ao2}
Production capacity at recycling plant a for product o3	λa_{3ao3}
Production capacity at recycling plant b	λB_b
The rate of poultry waste production by chicken rearing facility i	γ_i
The rate of poultry waste type 1 production by slaughterhouse j	φ_{1j}
The rate of poultry waste type 2 production by slaughterhouse j	φ_{2j}
The rate of poultry waste production by distributor m	α_m
The rate of product waste production by processing factory l	β_l
The rate of recycled product production by recycling factory n for product s1	εns_{1n}
The rate of recycled product production by recycling factory n for product s2	εns_{2n}
The rate of recycled product production by recycling factory n for product s3	εns_{3n}
The rate of recycled product production by recycling factory a for product a1	εao_{1a}
The rate of recycled product production by recycling factory a for product a2	εao_{2a}
The rate of recycled product production by recycling factory a for product a3	εao_{3a}
The rate of recycled product production by recycling factory b for product t	εb_b
Demand for final poultry products	Dp_{0p0}
Demand for poultry meal product by customer p1	Dp_{1p1}
Demand for dried protein products by customer p2	Dp_{2p2}
Demand for oil and margarine products by customer p3	Dp_{3p3}
Demand for biomass products by customer p4	Dp_{4p4}
Demand for organic fertilizer products by customer p5	Dp_{5p5}
Demand for compost product by customer p6	Dp_{6p6}
Demand for feather products by customer p7	Dp_{7p7}

Amount of product transported from chicken farm i to slaughterhouse j	X_{ij}
Decision variable	
Amount of product transported from slaughterhouse j to wholesale market k	U_{jk}
Amount of product transported from slaughterhouse j to processing factory l	S_{jl}
Amount of product transported from wholesale market k to distribution center m	W_{km}
Amount of product transported from processing factory l to distribution center m	V_{lm}
Amount of product transported from processing factory l to wholesale market k	gl_k
Amount of product transported from distribution center m to market p_0	mp_{mp_0}
Amount of poultry waste transferred from slaughterhouse j to recycling factory n	y_{jn}
Amount of poultry waste transferred from distributor m to recycling factory n	p_{mn}
Amount of poultry waste transferred from processing factory l to recycling factory n	G'_{ln}
Amount of poultry waste transferred from chicken farm i to recycling factory a	e_{ia}
Amount of poultry waste transferred from slaughterhouse j to comprehensive processing factory b	f_{jb}
Amount of products transferred from recycling factory n to market p_1	un_{np_1}
Amount of products transferred from recycling factory n to customer market p_2	kn_{np_2}
Amount of products transferred from recycling factory n to customer market p_3	pn_{np_3}
Amount of products transferred from recycling factory a to customer market p_4	xa_{ap_4}
Amount of products transferred from recycling factory a to customer market p_5	fa_{ap_5}
Amount of products transferred from recycling factory a to customer market p_6	wa_{ap_6}
Amount of products transferred from recycling factory b to customer market p_7	sb_{bp_7}
If chicken farm i is opened in the selected location, it equals 1; otherwise, it is 0	mr_i
If slaughterhouse j is opened in the selected location, it equals 1; otherwise, it is 0	Di_j
If wholesale market k is opened in the selected location, it equals 1; otherwise, it is 0	Wh_k
If processing factory l is opened in the selected location, it equals 1; otherwise, it is 0	Fr_l
If recycling factory n is opened in the selected location, it equals 1; otherwise, it is 0	Wp_n
If recycling factory a is opened in the selected location, it equals 1; otherwise, it is 0	wp'_a
If recycling factory b is opened in the selected location, it equals 1; otherwise, it is 0	wp''_b

The objective function of the mentioned model consists of three components:

- Establishment cost of centers.
- Transportation cost between centers.
- Processing cost of products at the centers.

Mathematical formulation of the problem

$$\begin{aligned}
 & \text{minimize } Z_1 + Z_2 + Z_3 \\
 & Z_1 = \sum_{i=1}^I fc_i \times Mr_i + \sum_{j=1}^J fk_j \times di_j + \\
 & \sum_{k=1}^K fm_k \times wh_k + \sum_{l=1}^L fl_l \times Fr_l + \\
 & \sum_{n=1}^N fj_n \times Wp_n + \sum_{a=1}^A fa_a \times wp'_a + \\
 & \sum_{b=1}^B fb_b \times wp''_b.
 \end{aligned} \tag{1}$$

$$\begin{aligned}
 Z_2 = & \sum_{i=1}^I \sum_{j=1}^J cx_{ij} \times X_{ij} + \\
 & \sum_{j=1}^J \sum_{k=1}^K cu_{jk} \times U_{jk} + \sum_{j=1}^J \\
 & \sum_{l=1}^L ca_{jl} \times S_{jl} + \sum_{k=1}^K \\
 & \sum_{m=1}^M cb_{km} \times W_{km} + \sum_{l=1}^L \\
 & \sum_{k=1}^K cs_{lk} \times g_{lk} + \sum_{l=1}^L \\
 & \sum_{m=1}^M ch_{lm} \times V_{lm} + \sum_{j=1}^J \\
 & \sum_{n=1}^N cp_{jn} \times y_{jn} + \sum_{m=1}^M \\
 & \sum_{n=1}^N ck_{mn} \times p_{mn} + \sum_{l=1}^L \\
 & \sum_{n=1}^N cd_{ln} \times G_{ln} + \sum_{i=1}^I \\
 & \sum_{a=1}^A cj_{ia} \times e_{ia} + \sum_{j=1}^J \\
 & \sum_{b=1}^B cm_{jb} \times f_{jb} + \sum_{p1=1}^{P1} \\
 & \sum_{n=1}^N cls_{1np1} \times un_{np1} + \\
 & \sum_{p2=1}^{P2} \sum_{n=1}^N cls_{2np2} \times pn_{np2} + \\
 & \sum_{p3=1}^{P3} \sum_{n=1}^N cls_{3np3} \times kn_{np3} + \\
 & \sum_{a=1}^A \sum_{p4=1}^{P4} clo_{1ap4} \times xa_{ap4} + \\
 & \sum_{a=1}^A \sum_{p5=1}^{P5} clo_{2ap5} \times fa_{ap5} + \\
 & \sum_{a=1}^A \sum_{p6=1}^{P6} clo_{3ap6} \times wa_{ap6} + \sum_{b=1}^B \\
 & \sum_{p7=1}^{P7} clt_{bp7} \times sb_{bp7} + \\
 & \sum_{m=1}^M \sum_{p0=1}^{P0} cp0_{mp0} \times mp0_{mp0}
 \end{aligned} \tag{2}$$

$$\begin{aligned}
 Z_3 = & \sum_{i=1}^I \sum_{j=1}^J cpi_i \times X_{ij} + \\
 & \sum_{j=1}^J cpj_j (\sum_{k=1}^K U_{jk} + \sum_{l=1}^L S_{jl}) + \\
 & \sum_{l=1}^L cpl_l (\sum_{k=1}^K g_{lk} + \sum_{m=1}^M V_{lm}) + \\
 & \sum_{k=1}^K \sum_{m=1}^M cpk_k \times W_{km} + \sum_{po=1}^{P0} \\
 & \sum_{m=1}^M cpm_m \times mp_{mp0} + \sum_{n=1}^N \\
 & \sum_{p1=1}^{P1} cpns1_n \times un_{np1} + \sum_{n=1}^N \\
 & \sum_{p2=1}^{P2} cpns2_n \times pn_{np2} + \sum_{n=1}^N \\
 & \sum_{p3=1}^{P3} cpns3_n \times kn_{np3} + \sum_{a=1}^A \\
 & \sum_{p4=1}^{P4} cpao1_a \times xa_{ap4} + \sum_{a=1}^A \\
 & \sum_{p5=1}^{P5} cpao2_a \times fa_{ap5} + \sum_{a=1}^A \\
 & \sum_{p6=1}^{P6} cpao3_a \times wa_{ap6} + \sum_{b=1}^B \\
 & \sum_{p7=1}^{P7} cpb_b \times sb_{bp7}.
 \end{aligned} \tag{3}$$

The constraints of the mathematical model These constraints are formulated so that the network should strive to operate within specified capacities, demands, and logistics limitations at every stage of the CLSC. Such constraints help balance supply and demand, followed by the capacities of facilities for staging and storage, apart from maintaining the network operations in such a way that they are cost-effective. The limitations are combined to arrive at an optimum configuration of CLSC with minimum costs to satisfy all operational requirements regarding establishment, transportation, and processing costs across the network.

Table 2 The summary of the related paper

Authors	Year	Network		Objective function		Model type			Solution approaches		Object(s)	Case
		CLSC	SC	Multi	Single	MILP	MINLP	Other	Exact	Metaheuristics		
Hassani et al	2011	✓		✓		✓			✓		Minimizing the total cost	Perishable goods
Khalilpourazari & Arshadi khamseh	2019		✓	✓		✓			✓		Minimizing the total cost	Blood
Yavari & Geraeli	2019	✓		✓		✓			✓		Minimizing transportation time Minimizing the total cost Minimizing environmental impact	Dairy
Zandkarimkhani et al	2020	✓		✓		✓			✓		Minimizing the total cost	Medicine
Hamdan & Diabat	2020		✓	✓					✓		Minimizing the total cost	Blood
Kaya & Ozkok	2020	✓			✓		✓		✓		Minimizing transportation time	blood
Jouzdani & Govindan	2021		✓	✓					✓		Minimizing the total cost	Dairy
Mosallanezhad et al	2021	✓			✓	✓				✓	Minimizing the total cost	shrimp
Seyfi-Shishavan et al	2021		✓	✓		✓			✓		Minimizing the total cost	blood
Fallahi et al	2021	✓			✓	✓			✓		Minimizing the total cost	blood
Yang & Chen	2021	✓		✓			✓		✓		Minimizing the total cost	Electronic devices
Karadağ et al	2021	✓		✓		✓			✓		Maximizing the job opportunities	blood
Baghizadeh et al	2021	✓		✓		✓			✓		Minimizing transportation time Minimizing the total cos Maximizing social scales	Reproduction
Salehi-Amiri et al	2021(b)	✓			✓	✓			✓		Minimizing the total cost	Avocado
Tavana et al	2022	✓		✓		✓			✓		Minimizing the total cost Minimizing co2 product Maximizing the job opportunities	-----
Fallahi et al	2022	✓		✓			✓			✓	Minimizing the total cost	Blood plasma

Table 2 (continued)

Authors	Year	Network		Objective function		Model type			Solution approaches		Object(s)	Case
		CLSC	SC	Multi	Single	MILP	MINLP	Other	Exact	Metaheuristics		
											Profit	Cost
Rajabi-Kafshgar et al. Abbasi et al.	2023	✓			✓	✓			✓		Minimizing co2 product	Pistachio
	2023	✓		✓		✓			✓		Minimizing the total cost Minimizing the total cost	Automotive industry
Gholipour et al.	2023	✓		✓		✓			✓		Minimizing co2 product Minimizing the total cost	pomegranate
	2023	✓		✓		✓			✓		Minimizing supply risk Minimizing the total cost	citrus
Keshavarz-Ghorbani & Pasandideh	2023	✓		✓			✓		✓		Minimizing environmental impact Related to social responsibility Maximizing total benefit	Mobile phone
	2024	✓			✓			✓			Maximizing social benefit Minimizing the economic and exergetic costs	Food
Khalili-Fard et al. Gharye Mirzaei et al.	2024	✓		✓				✓			Minimizing the total cost	-----
	2024	✓			✓				✓		Minimizing the total cost	pistachio
This work	2024	✓			✓		✓		✓		Minimizing the total cost	Poultry

$$\sum_{i=1}^I mr_i \geq 1 \quad (4)$$

$$\sum_{j=1}^J Di_j \geq 1 \quad (5)$$

$$\sum_{l=1}^L Fr_l \geq 1 \quad (6)$$

$$\sum_{k=1}^K Wh_k \geq 1 \quad (7)$$

$$\sum_{n=1}^N wp_n \geq 1 \quad (8)$$

$$\sum_{a=1}^A wp'_a \geq 1 \quad (9)$$

$$\sum_{b=1}^B wp''_b \geq 1 \quad (10)$$

$$\sum_{j=1}^J X_{ij} \leq \lambda I_i, \forall i \in I \quad (11)$$

$$\sum_{k=1}^K gl_k + \sum_{m=1}^M v_{lm} \leq \lambda L_l, \forall l \in L \quad (12)$$

$$\sum_{l=1}^L s_{jl} + \sum_{k=1}^K u_{jk} \leq \lambda J_j, \forall j \in J \quad (13)$$

$$\sum_{i=1}^I X_{ij} \leq \lambda J_j \times Di_j, \forall j \in J \quad (14)$$

$$\sum_{l=1}^L s_{jl} \leq \lambda L_l \times fr_l, \forall j \in J \quad (15)$$

$$\sum_{j=1}^J u_{jk} + \sum_{l=1}^L gl_k \leq \lambda K_k \times wh_k, \forall k \in K \quad (16)$$

$$\sum_{m=1}^M w_{km} \leq \sum_{j=1}^J u_{jk} + \sum_{l=1}^L g_{lk}, \forall k \in K \quad (17)$$

$$\sum_{k=1}^K w_{km} + \sum_{l=1}^L v_{lm} \leq \lambda M_m, \forall m \in M \quad (18)$$

$$\sum_{m=1}^M mp_{mp_0} \geq Dp_{0_{p_0}}, \forall p_0 \in P_0 \quad (19)$$

$$\sum_{a=1}^A e_{ia} \leq \sum_{j=1}^J \gamma_i \times \lambda I_i, \forall a \in A \quad (20)$$

$$\sum_{p_0=1}^{P_0} mp_{mp_0} \leq (1 - \alpha_m) \times \left(\sum_{l=1}^L v_{lm} + \sum_{k=1}^K w_{km} \right), \forall m \in M \quad (21)$$

$$\sum_{i=1}^I (1 - (\varphi_{1_j} + \varphi_{2_j})) \times x_{ij} \geq \sum_{l=1}^L s_{jl} + \sum_{k=1}^K u_{jk}, \forall j \in J \quad (22)$$

$$\sum_{j=1}^J (1 - \beta_l) \times s_{jl} \geq \sum_{k=1}^K g_{lk} + \sum_{m=1}^M v_{lm}, \forall l \in L \quad (23)$$

$$\sum_{j=1}^J y_{jn} + \sum_{l=1}^L g'_{ln} + \sum_{m=1}^M p_{mn} \leq \left[\sum_{s1=1}^{S1} \lambda n_{1ns1} + \sum_{s2=1}^{S2} \lambda n_{2ns2} + \sum_{s3=1}^{S3} \lambda n_{3ns3} \right] \times wp_n, \forall n \in N \quad (24)$$

$$\sum_{i=1}^I e_{ia} \leq \left(\sum_{o1}^{O1} \lambda a_{1ao1} + \sum_{o2}^{O2} \lambda a_{2ao2} + \sum_{o3}^{O3} \lambda a_{3ao3} \right) \times wp'_a, \forall a \in A \quad (25)$$

$$\sum_{j=1}^J f_{jb} \leq \lambda B_b \times wp''_b, \forall b \in B \quad (26)$$

$$\sum_{j=1}^J (\beta_l) \times S_{jl} \geq \sum_{l=1}^L G_{ln}, \forall n \in N \quad (27)$$

$$\alpha_m \times \left(\sum_{l=1}^L v_{lm} + \sum_{k=1}^K w_{km} \right) \geq \sum_{n=1}^N p_{mn}, \forall m \in M \quad (28)$$

$$\sum_{i=1}^I \varphi 2_j \times x_{ij} \geq \sum_{b=1}^B f_{jb}, \forall j \in J \quad (29)$$

$$\sum_{l=1}^L G_{ln} + \sum_{j=1}^J Y_{jn} + \sum_{m=1}^M p_{mn} \geq \sum_{p_1=1}^{P_1} \frac{un_{np_1}}{\varepsilon n s_{1n}} + \sum_{p_2=1}^{P_2} \frac{kn_{np_2}}{\varepsilon n s_{2n}} + \sum_{p_3=1}^{P_3} \frac{pn_{np_3}}{\varepsilon n s_{3n}} + \frac{\sum_{p_5=1}^{P_5} f_{a_{ap_5}}}{\varepsilon a o_{2a}}, \forall n \in N \quad (30)$$

$$\sum_{i=1}^I e_{ia} \geq \sum_{p_4=1}^{P_4} \frac{x a_{ap_4}}{\varepsilon a o_{1a}} + \sum_{p_5=1}^{P_5} \frac{f a_{ap_5}}{\varepsilon a o_{2a}} + \sum_{p_6=1}^{P_6} \frac{w a_{ap_6}}{\varepsilon a o_{3a}}, \forall a \in A \quad (31)$$

$$\sum_{j=1}^J f_{jb} \times \varepsilon b_b \geq \sum_{p_7=1}^{P_7} s b_{bp_7}, \forall b \in B \quad (32)$$

$$\sum_{n=1}^N un_{np_1} \geq D p_{1p_1}, \forall p_1 \in P_1 \quad (33)$$

$$\sum_{n=1}^N kn_{np_2} \geq D p_{2p_2}, \forall p_2 \in P_2 \quad (34)$$

$$\sum_{n=1}^N pn_{np_3} \geq D p_{3p_3}, \forall p_3 \in P_3 \quad (35)$$

$$\sum_{a=1}^A x a_{ap_4} \geq D p_{4p_4}, \forall p_4 \in P_4 \quad (36)$$

$$\sum_{a=1}^A f a_{ap_5} \geq D p_{5p_5}, \forall p_5 \in P_5 \quad (37)$$

$$\sum_{a=1}^A w a_{ap_6} \geq D p_{6p_6}, \forall p_6 \in P_6 \quad (38)$$

$$\sum_{b=1}^B s b_{bp_7} \geq D p_{7p_7}, \forall p_7 \in P_7 \quad (39)$$

$$mr_i Di_j, wh_k, fr_l, wp_n, wp'_a, wp''_b \in \{0,1\}, \forall i \in I, \forall j \in J, \forall k \in K, \forall l \in L, \forall n \in N, \forall a \in A, \forall b \in B \quad (40)$$

$$x_{ij}, u_{jk}, s_{jl}, w_{km}, v_{lm}, g_{lk}, y_{jn}, p_{mn}, G_{ln}, e_{ia}, f_{jb}, un_{np_1}, kn_{np_2}, pn_{np_3}, x a_{ap_4}, f a_{ap_5}, w a_{ap_6}, s b_{bp_7} \geq 0. \quad (41)$$

The objective function of the proposed PSC is to minimize the comprehensive chain cost, encompassing the establishment and opening costs of various centers, transportation costs between them, and the processing costs incurred at the centers. The objective function is mathematically defined by Eq. 1, Eq. 2, and Eq. 3. Now, a set of constraints has been developed to ensure the proper functionality of SC. Constraints (4–10) ensure that at least one center should be opened. Constraints (11–16) restrict the opened centers' capacities. Constraints (17–23) balance the input and output flow at the centers.

Similarly, Constraints (24–26) regulate the capacities of the recycling plants, while Constraints (27–32) are responsible for the Flow balance between input and output through the recycling plants. Constraints (33–39) guarantee that the market demand regarding recycled products for customers p_1 to p_7 is satisfied. Constraint (40) imposes binary variables as an indication of certain conditions as 0 or 1. Constraint (41) ensures non-negativity for continuous decision variables. Figure 3 model represents several inter-layer links within the PSC. It also represents upward and downward connections of chicken and bird farms with slaughterhouses, processing factories, wholesale centers, distribution centers, poultry waste recycling plants, and customer markets for recycled products. Given that the model size grows with the number of entities involved, the paper's authors propose using metaheuristic algorithms to solve real-world instances of the PSC problem in a reasonable time.

3.2 Problem-solving method

As mentioned before, real-world SC problems are intrinsically complex and often NP-hard. Applying exact mathematical methods to such problems, especially large ones, is usually time-consuming and inefficient.

Over the past few years, researchers have applied many metaheuristic approaches to find appropriate solutions to solve NP-hard problems. Time savings are often beneficial in most cases. The primary advantages of these methods include handling complex problems and avoiding local optimality (Fathollahi-Fard et al., 2020; Seyedi & Maleki-Daronkolaei, 2013; Navaei et al., 2010). For example, in addressing the order acceptance and SC scheduling problem, researchers applied Genetic Algorithm and variable neighborhood search (Govindan et al., 2015). In this research, five metaheuristic approaches are applied: Genetic Algorithm (GA), Simulated Annealing Algorithm (SA), Differential Evolution Algorithm (DE), and two hybrid methods - Hybrid Genetic Algorithm with Simulated Annealing (HGASA) and Hybrid Differential Evolution and Simulated Annealing Algorithm (HDESA). These hybrid approaches combine the strengths of individual algorithms to achieve improved solution quality in the PSC network optimization. The following sections discuss each algorithm and provide pseudocode for hybrid algorithms.

3.2.1 Genetic algorithm

One of the most well-known approaches in evolutionary algorithms is the GA, which will help find successful solutions in various applications for various areas. The idea behind the GA was proposed by Holland (1992). The development was inspired by genetics and natural selection. GA applies two leading operators to make this algorithm's search process more intensive and diverse. Besides these two operators, a probabilistic selection can be used to make proper choices within the algorithm: crossover and mutation.

3.2.2 Simulated annealing algorithm (SA)

The SA was proposed by Kirkpatrick et al. (1983) and inspired by annealing in metallurgy. It is a local search strategy for optimization problems. Starting from an initial solution, SA generates neighbors iteratively and accepts a better neighbor or, with a probability of $\exp(-\Delta E/T)$, a worse neighbor. This process is repeated for different temperatures, gradually lowering the temperature.

3.2.3 Hybrid genetic algorithm with simulated annealing (HGASA)

GA performs a better identification of regions containing the global and local optima within the problem state space. However, they experience their instability issues at later stages. One of the safer meta-heuristic approaches is to combine several techniques in the process of problem-solving. The pseudocode for the hybrid genetic algorithm with SA is presented in Fig. 4. Recently, the interest of many researchers has shifted toward evolutionary algorithms. However, hybrid algorithms can overcome these shortcomings to explore a more

```

Population: Generate initial random Population
Cost: Evaluate initial Population
Maxit: Maximum iteration
N-crossover: Number of off springs
N-mutation: Number of mutation
for it=1: Maxit
    Select Parents for offspring Production
    for each N-crossover
        Do crossover
    end for
    for each N-mutation
        Do SA
    end for
    Evaluate Population
    Update the Best individual
End for
Returning the Best individual

```

Fig. 4 Hybrid genetic SA algorithm pseudocode

comprehensive solution space. The combination of algorithms offers the opportunity to strike a better balance between these crucial stages, having higher expectations for achieving better solutions. The hybrid GA and SA can enhance solution efficiency. SA is considered a local search operator capable of accepting or rejecting new local solutions. Hence, this research employs a combination of GA with SA.

3.2.4 Differential evolution algorithm (DE)

The DE algorithm is a valuable metaheuristic for optimizing large-scale problems, which are often challenging to solve using analytical methods within a reasonable time frame. Its

```

Input: NP: the population size
K: the temperature reduction parameter
G: the maximum number of iteration
T: the current temperature
M: the number of SA iteration
1: generate an initial population sampling from the feasible
   Solution space randomly
2: calculate the fitness value of the individuals
3: for g=1 to G do
4:   for l=1 to m do
5:     perform mutation operation as shown in Algorithm 1
6:   end for
7:   for j = 1 to D do
8:     j rand = rndreal (1, D)
9:     if rndreal (0,1) < cr or j ==j rand then
10:    ui,j = xr1,j +F.(xr2,j − xr3,j)
11:   else
12:    ui,j = xi1,j
13:   end if
14: end for
15: for i= 1 NP do
16:   perform selection operation as shown in Algorithm 2
17: end for
18: T=K×T
19: end for

```

Fig. 5 Hybrid DE and SA algorithm pseudocode

Table 3 Proposed chromosome schematic design

S.1	S.2				S.3			
M	P0		N	P1		N	P2	
0.110934	0.398689	0.98587	0.442547	0.50004	0.634964	0.200842	0.140453	0.036509
S.4			S.5			S.6		
N	P3		A	P4		A	P5	
0.117518	0.820774	0.441895	0.515387	0.685695	0.696263	0.256497	0.114679	0.668804
S.7			S.8			S.9		
A	P6		B	P7		F	B	
0.014332	0.043808	0.853135	0.16946	0.502641	0.777938	0.856	0.123	0.059
S.10			S.11					
L	A		K+L			M		
0.088	0.391	0.471	0.792	0.377	0.340	0.340	0.588	
S.12						S.13		
J+L			K			J	L	
0.482	0.302	0.270	0.415	0.332	0.795	0.206715	0.536269	0.647529
S.14			S.15					
L	J		J+L+M				N	
0.777602	0.419726	0.218543	0.06304	0.957979	0.681425	0.849914	0.05347	0.099357
S.16								
(J+L+M)*N								
0.22868	0.378491	0.800403	0.336108	0.68613	0.797477	0.284281	0.90416	0.466656
		0.615323	0.142257					0.524224

Table 4 Priority-based chromosome and random key

S.1			
M		P0	
1	2	2	1

efficiency, simplicity, and rapid convergence make it a popular choice among researchers tackling complex problems (Mokhtari & Mozdgir, 2015).

3.2.5 Hybrid differential evolution and simulated annealing algorithm (HDESA)

DE is one of the best algorithms for global optimization. However, traditional DE has relatively poor search performance and often converges to a local optimum. Given this improvement, a new variant of DE, Hybrid DE, is combined with the SA algorithm, thereby forming the Hybrid DE and SA-HDESA algorithm, as proposed in the paper under reference. It incorporates the concept of ranking into the DE mutation process and injects the SA concept into the selection operations. The former type generally aims at enhancing the exploitation abilities and raising the efficiency in search, while the latter generally aims at enhancing the exploration capabilities and preventing the algorithm from converging into local optima. The pseudocode of the algorithm can be seen in Fig. 5.

3.3 Encryption and decryption

A random key method was utilized for encoding. In this manner, an encoded solution comprises a vector consisting of several random numbers between zero and one. These random numbers indicate the priority of assigning a value to a variable or the extent of value allocation to a variable. Any vector of this form will be mapped to a feasible solution for the mathematical model of the problem. Taking as example a case where the size of each set in the mathematical model is 2, Table 3 displays a sample of a randomly encoded solution for that example, divided into 16 sectors for convenience, S.1 through S.16.

Except for section number 16, all sections of this answer indicate the priority of centers for variable allocation, wherein a smaller randomly assigned number signifies a higher priority for the center. For instance, to determine the variable value of mp_mp0 , it should first be established which of the market p_0 's demands. Based on the random numbers provided in the above figure, initially, the second market p_0 and then the first market is chosen for this purpose. Subsequently, it is determined which center m is used first to supply goods to the markets. In summary, the centers' priority for assigning values to the mp 's variable is determined according to Table 4.

Then, the allocation of variables is carried out considering the minimum demand and the capacity of supply centers to ensure the feasibility of the solution. For instance, for the above figure, initially, the variable value mp_{12} is determined as $mp_{12} = \min\{Dp_{02}, \lambda M_1\}$. Such allocations continue until the demand is met or the center's capacity is exhausted. A similar process is followed for the encoded response sections (except Sect. 16).

Section 16, the encoded response, signifies the allocation value of variables y_{jn} , p_{mn} , G_{ln} . It is essential to note that the priority for allocating these variables is determined in Sect. 15. After establishing the priority, an equation such as $x = rand * x_{max}$ is used to allocate values to these variables, where x represents the variable value, $rand$ denotes the corresponding random number in the encoded response, and x_{max} is the maximum value

Table 5 Designed test problems for the model

Indices																	
Classification	Problem number	i	j	k	l	m	n	a	b	$p0$	$p1$	$p2$	$p3$	$p4$	$p5$	$p6$	$p7$
Small	1	2	2	1	1	2	1	1	2	2	2	2	2	2	2	2	2
	2	3	3	2	2	3	2	2	3	3	3	3	3	3	3	3	3
	3	5	5	3	4	5	4	4	5	4	5	4	5	4	4	4	4
	4	7	7	4	6	7	6	6	7	5	7	6	7	5	5	5	6
	5	10	10	5	9	10	9	8	10	6	8	8	8	6	6	6	8
	6	13	13	6	12	13	12	10	13	8	9	10	9	8	8	8	10
	7	16	16	7	15	16	15	13	16	10	11	12	11	10	10	10	12
Medium	8	21	19	8	18	19	18	16	19	12	13	14	13	12	12	12	14
	9	24	22	9	21	22	21	20	22	14	16	16	16	15	15	15	16
	10	29	25	10	24	25	24	24	25	16	19	18	19	18	18	18	18
	11	35	31	11	30	31	30	28	31	18	22	20	22	20	20	20	21
	12	40	37	12	36	37	36	32	37	20	25	22	25	24	24	24	24
	13	45	43	13	42	43	42	36	43	24	30	24	30	28	28	28	27
	14	50	49	14	48	49	48	40	49	28	35	26	35	32	32	32	30
Large	15	60	55	15	54	55	54	45	55	32	45	28	45	36	36	36	33
	16	70	70	16	69	70	60	50	70	36	55	30	55	42	42	42	36
	17	90	85	17	84	85	75	55	85	40	75	33	75	50	50	50	39
	18	110	100	18	99	100	90	60	100	44	95	36	95	58	58	58	42
	19	130	115	19	114	115	105	70	115	48	110	39	110	66	66	66	45
	20	150	130	20	129	130	120	80	130	52	125	42	125	80	80	80	48
	21	170	150	21	149	150	140	90	150	56	140	45	140	90	90	90	51
	22	190	170	22	169	170	160	100	170	60	155	48	155	100	100	100	54
	23	210	190	23	189	190	180	110	190	70	170	51	170	110	110	110	57
	24	230	210	24	209	210	200	120	210	80	185	55	185	120	120	120	60

Table 6 Various values of model parameters

Parameters	Amount	Unit	Parameters	Amount	Unit
f_{c_i}	uniform $\sim [10, 30]$	Dollar (\$)	$ds1_{np1}$	uniform $\sim [40, 50]$	Dollar (\$) per Ton
f_{k_j}	uniform $\sim [20, 45]$	Dollar (\$)	$ds2_{np2}$	uniform $\sim [40, 50]$	Dollar (\$) per Ton
f_{b_l}	uniform $\sim [15, 35]$	Dollar (\$) per Ton	$ds3_{np3}$	uniform $\sim [40, 50]$	Dollar (\$) per Ton
f_{m_k}	uniform $\sim [10, 20]$	Dollar (\$) per Ton	$do1_{ap4}$	uniform $\sim [40, 50]$	Dollar (\$) per Ton
f_{j_n}	uniform $\sim [10, 25]$	Dollar (\$) per Ton	$do2_{ap5}$	uniform $\sim [40, 50]$	Dollar (\$) per Ton
f_{l_a}	uniform $\sim [10, 25]$	Dollar (\$) per Ton	$do3_{ap6}$	uniform $\sim [40, 50]$	Dollar (\$) per Ton
f_{v_b}	uniform $\sim [10, 25]$	Dollar (\$) per Ton	dt_{bp7}	uniform $\sim [40, 50]$	Dollar (\$) per Ton
cx_{ij}	uniform $\sim [80, 110]$	Dollar (\$) per Ton	λI_i	[500]	Dollar (\$) per Ton
cu_{jk}	uniform $\sim [55, 75]$	Dollar (\$) per Ton	λJ_j	[500]	Dollar (\$) per Ton
ca_{jl}	uniform $\sim [45, 58]$	Dollar (\$) per Ton	λL_l	[200]	Dollar (\$) per Ton
cb_{km}	uniform $\sim [62, 80]$	Dollar (\$) per Ton	λK_k	[300]	Dollar (\$) per Ton
cs_{lk}	uniform $\sim [30, 70]$	Dollar (\$) per Ton	λM_m	[400]	Dollar (\$) per Ton
co_{lm}	uniform $\sim [35, 45]$	Dollar (\$) per Ton	$\lambda n1_{ns1}$	uniform $\sim [100, 200]$	Dollar (\$) per Ton
cp_{jn}	uniform $\sim [35, 45]$	Dollar (\$) per Ton	$\lambda n2_{ns2}$	uniform $\sim [100, 200]$	Tons
cd_{ln}	uniform $\sim [25, 40]$	Dollar (\$) per Ton	$\lambda n3_{ns3}$	uniform $\sim [100, 200]$	Tons
ck_{mn}	uniform $\sim [25, 40]$	Dollar (\$) per Ton	$\lambda a1_{ao1}$	uniform $\sim [100, 200]$	Tons
cj_{ia}	uniform $\sim [25, 40]$	Dollar (\$) per Ton	$\lambda a2_{ao2}$	uniform $\sim [100, 200]$	Tons
cm_{jb}	uniform $\sim [40, 50]$	Dollar (\$) per Ton	$\lambda a3_{ao3}$	uniform $\sim [100, 200]$	Tons
γ_i	[0.5, 0.5]	Dollar (\$) per Ton	λB_b	uniform $\sim [100, 300]$	Tons
φ_j	[0.1, 0.1]	Tons	εb_b	[0.90, 0.85, 0.95]	percentage
α_m	[0.1, 0.1]	percentage	Dv_l	uniform $\sim [150, 250]$	Tons
β_l	[0.1, 0.1]	percentage	Dp_k	uniform $\sim [150, 250]$	Tons
$\varepsilon ns1_n$	[0.9, 0.85, 0.95]	percentage	Dk_m	uniform $\sim [750, 850]$	Tons
$\varepsilon ns2_n$	[0.75, 0.850, 90]	percentage	$Dp1_{p1}$	uniform $\sim [30, 60]$	Tons
$\varepsilon ns3_n$	[0.65, 0.60, 0.75]	percentage	$Dp2_{p2}$	uniform $\sim [30, 60]$	Tons
$\varepsilon ao1_a$	[0.95, 0.90, 0.85]	percentage	$Dp3_{p3}$	uniform $\sim [30, 60]$	Tons

Table 6 (continued)

Parameters	Amount	Unit	Parameters	Amount	Unit
$\varepsilon a o_{2a}$	[0.75, 0.85, 0.80]	percentage	Dp_{4p4}	uniform $\sim$ [30,60]	Tons
$\varepsilon a o_{3a}$	[0.85, 0.80, 0.90]	percentage	Dp_{5p5}	uniform $\sim$ [30,60]	Tons
γ_i	[0.5, 0.5]	Tons	Dp_k	uniform $\sim$ [150,250]	Tons
φ_j	[0.1, 0.1]	Tons	Dk_m	uniform $\sim$ [750,850]	Tons
α_m	[0.1, 0.1]	percentage	Dp_{1p1}	uniform $\sim$ [30,60]	Tons
β_l	[0.1, 0.1]	percentage	Dp_{2p2}	uniform $\sim$ [30,60]	Tons
$\varepsilon n s_{1n}$	[0.9, 0.85, 0.95]	percentage	Dp_{3p3}	uniform $\sim$ [30,60]	Tons

Table 7 Levels of metaheuristic parameters

Algorithm	Parameter	Description	Levels
SA	<i>Maxit</i>	Number of outer loop iterations	[160 180 200 220 240]
	<i>MaxSubit</i>	Number of inner loop iterations	[6 8 10 12 14]
	<i>T0</i>	Initial temperature	[100 200 300 400 500] *1000
	<i>alpha</i>	Cooling rate	[0.95 0.96 0.97 0.98 0.99]
GA	<i>Maxit</i>	Number of iterations (generations)	[160 180 200 220 240]
	<i>npop</i>	Number of chromosomes per population	[5 10 15 20 25]
	<i>PMutation</i>	Mutation rate	[0.15 0.2 0.25 0.3 0.35]
	<i>PCrossover</i>	Crossover rate	[0.15 0.2 0.25 0.3 0.35]
Hybrid GA and SA (GASA)	<i>Maxit</i>	Number of iterations (generations)	[160 180 200 220 240]
	<i>npop</i>	Number of chromosomes per population	[5 10 15 20 25]
	<i>PMutation</i>	Mutation rate	[0.15 0.2 0.25 0.3 0.35]
	<i>PCrossover</i>	Crossover rate	[0.15 0.2 0.25 0.3 0.35]
	<i>T0</i>	Initial temperature	[100 200 300 400 500] *1000
	<i>alpha</i>	Cooling rate	[0.95 0.96 0.97 0.98 0.99]
DE	<i>Maxit</i>	Number of iterations (generations)	[160 180 200 220 240]
	<i>npop</i>	Number of chromosomes per population	[5 10 15 20 25]
	<i>F</i>	Mutation coefficient	[0.6 0.7 0.8 0.9 1]
	<i>CR</i>	Crossover probability	[0.15 0.2 0.25 0.3 0.35]
DESA	<i>Maxit</i>	Number of iterations (generations)	[160 180 200 220 240]
	<i>npop</i>	Number of chromosomes per population	[6 8 10 12 14]
	<i>F</i>	Mutation coefficient	[0.6 0.7 0.8 0.9 1]
	<i>CR</i>	Crossover probability	[0.15 0.2 0.25 0.3 0.35]
	<i>T0</i>	Initial temperature	[100 200 300 400 500] *1000
	<i>alpha</i>	Cooling rate	[0.95 0.96 0.97 0.98 0.99]

Table 8 Parameter settings

Algorithm	Parameter	Small dimensions	Medium dimensions	Large dimensions
SA	<i>Maxit</i>	240	240	220
	<i>MaxSubit</i>	14	10	6
	<i>T0</i>	100000	200000	100000
	<i>alpha</i>	0.95	0.98	0.98
GA	<i>Maxit</i>	240	200	160
	<i>Npop</i>	20	25	25
	<i>PMutation</i>	0.35	0.3	0.3
	<i>PCrossover</i>	0.3	0.3	0.3
GASA	<i>Maxit</i>	200	180	220
	<i>Npop</i>	25	25	25
	<i>PMutation</i>	0.3	0.3	0.2
	<i>PCrossover</i>	0.35	0.35	0.3
	<i>T0</i>	200000	500000	200000
	<i>alpha</i>	0.99	0.98	0.96
DE	<i>Maxit</i>	180	160	220
	<i>npop</i>	25	25	25
	<i>F</i>	0.9	0.6	0.7
	<i>CR</i>	0.15	0.35	0.35
DESA	<i>Maxit</i>	160	200	200
	<i>npop</i>	14	14	12
	<i>F</i>	0.6	0.8	1
	<i>CR</i>	0.3	0.35	0.35
	<i>T0</i>	300000	300000	200000
	<i>alpha</i>	0.96	0.95	0.99

that the variable x can assume. This value equals the minimum between the remaining demand and capacity of the centers. This allocation is performed once, and if the demand is not met, a similar approach to Sects. 1 to 15 is employed. The reason for using this combined allocation method is that empirical experiments on the encoding method have shown better efficiency compared to a scenario where allocation is solely based on priority.

4 Analysis

4.1 Data generation

This paper has designed a series of test problems with various problem sizes to validate the developed model. In this context, 24 test problems have been crafted in small, medium, and large dimensions. Table 5 illustrates the test problems that have been created to fulfill the objectives of this study. We estimate the test problems using the parameters described in Table 6. Initially, the accuracy of the proposed model is examined using MATLAB coding software.

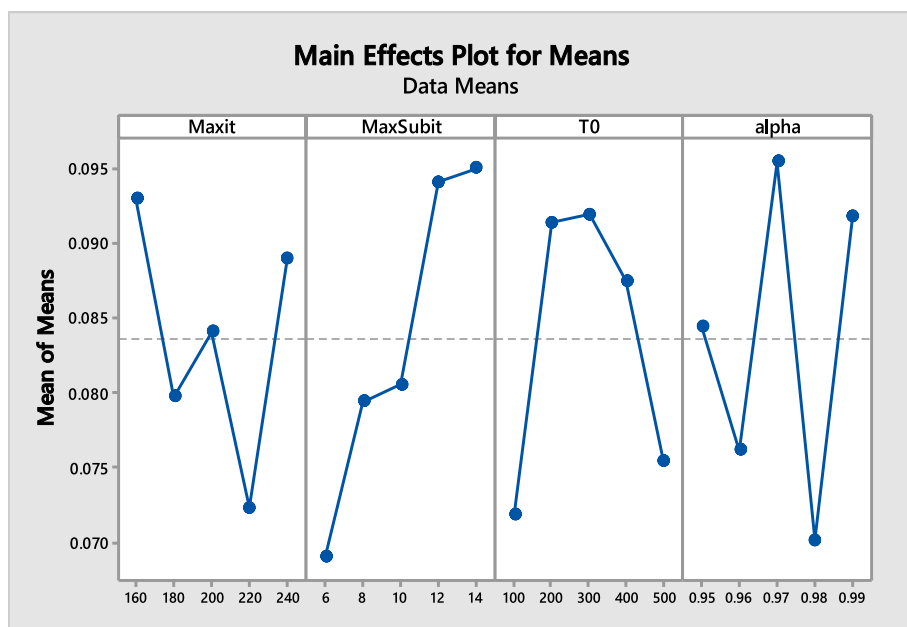


Fig. 6 SA algorithm parameter tuning in large-scale dimensions

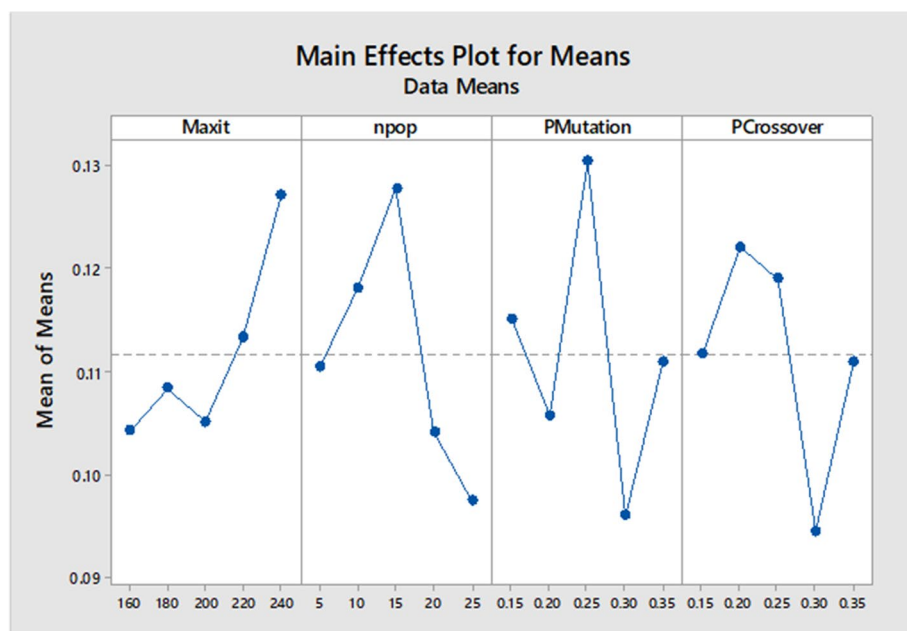


Fig. 7 GA algorithm parameter tuning in large-scale dimensions

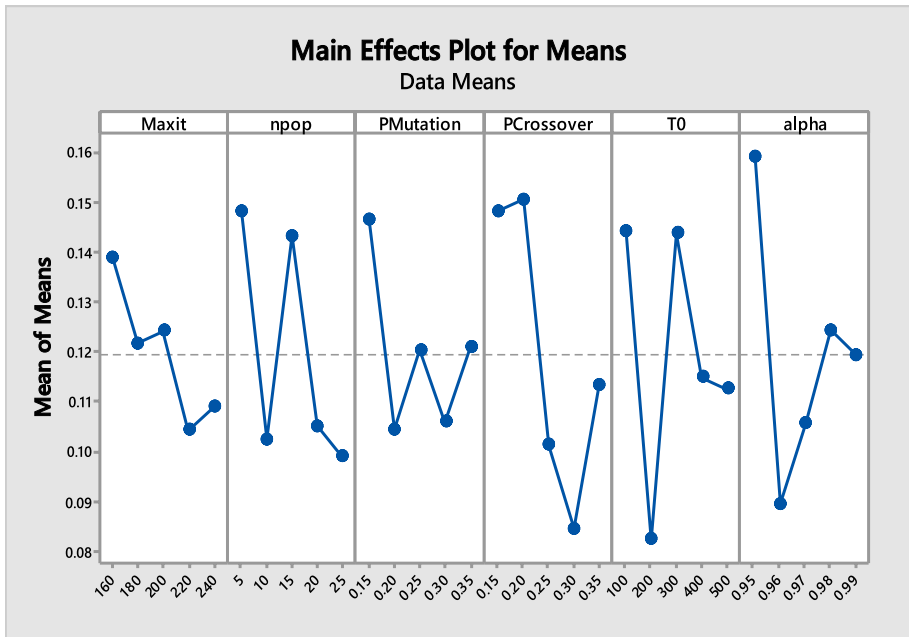


Fig. 8 GA-SA algorithm parameter tuning

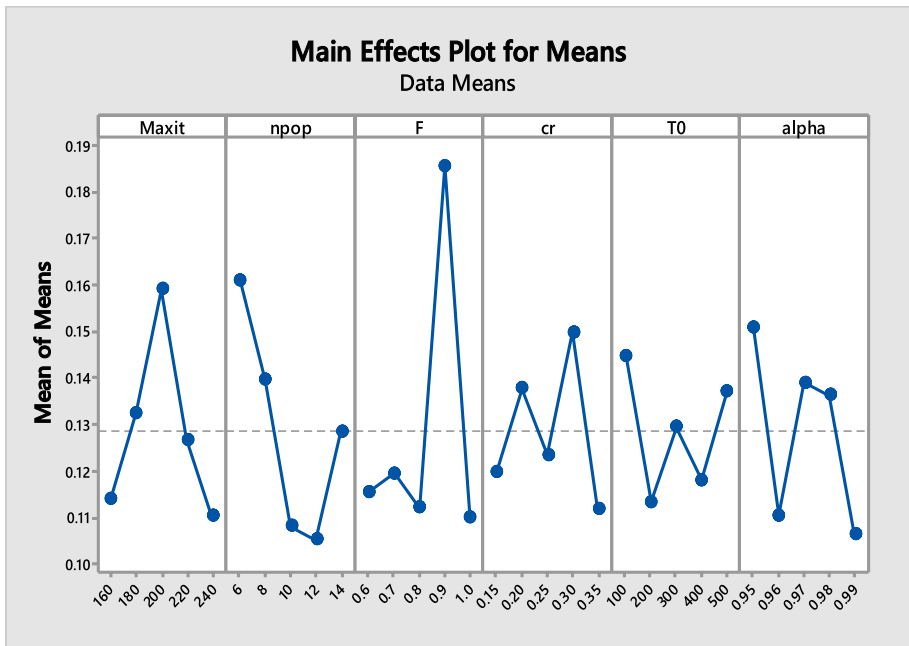


Fig. 9 DE algorithm parameter tuning

4.2 Parameter setting

This section employs the Taguchi method to outline the parameter configuration process for metaheuristic algorithms. Originally introduced by Taguchi (1986), the Taguchi method remains one of the most widely adopted robust optimization techniques in experimental design. Its primary objective is to identify an optimal parameter configuration that maximizes system performance and enhances robustness. The method operates by systematically varying input parameters across predefined levels and analyzing their effects on the output or performance metric. This structured approach allows researchers to efficiently determine the most influential factors while minimizing the number of required experiments, thereby optimizing computational efficiency (Hosseini Baboli et al., 2023; Seyed et al., 2022; Mozdgir et al., 2013).

With a view to the better clarity of the paper, a detailed specification of levels of parameters for metaheuristic algorithms was specified in Table 7, and the corresponding configured parameters were in Table 8. After that, the configuration of these parameters should apply the Taguchi method by taking into account different levels and combinations in a structured manner. The resultant configuration process can visually show the results obtained through the application of the Taguchi method, as represented in Figs. 6, 7, 8, 9 and 10. This will strengthen the efficiency and effectiveness of the metaheuristic algorithms in solving real-world instances of the PSC problem, which has been discussed in a broader context within the paper. The rest of the information, including the specific values for specific parameters, is given in Tables 7 and 8, Figs. 6, 7, 8, 9 and 10.



Fig. 10 DE-SA algorithm parameter tuning

Table 9 Impact of changes in recycled product production rates by poultry waste reprocessing plants on the objective function

Capacity parameter	Changes (tons)	Objective function value	Capacity parameter	Changes (tons)	Capacity
$\varepsilon ns1_n$	0.80	376555.202127	$\varepsilon ao1_a$	0.80	350075.011001
	0.85	370251.601209		0.85	349939.337788
	0.87	367950.397639		0.87	349889.480583
	0.90	364691.383155		0.90	348918.944184
	0.93	361640.801918		0.93	349752.958521
$\varepsilon ns2_n$	0.95	359713.956507	$\varepsilon ao2_a$	0.95	349711.283366
	0.80	366876.773663		0.80	350080.691959
	0.85	360312.869223		0.85	349948.118126
	0.87	367883.421483		0.87	349899.355449
	0.90	364441.562899		0.90	349830.274989
$\varepsilon ns3_n$	0.93	361221.596023	$\varepsilon ao3_a$	0.93	349765.651333
	0.95	359187.843711		0.95	349724.836393
	0.80	366144.065685		0.80	349956.200522
	0.85	359889.535237		0.85	349826.053463
	0.87	357589.128140		0.87	349778.183281
ε_b	0.90	354330.331861		0.90	349710.367189
	0.93	351281.912628		0.93	349646.926329
	0.95	349356.234806		0.95	349606.858417
	0.80	349880.199476			
	0.85	349791.964181			
	0.87	349759.509820			
	0.90	349713.532809			
	0.93	349670.522056			
	0.95	349643.357370			

4.3 Sensitivity analysis

Sensitivity analysis typically shows how changes in the input parameters affect the changed variables of the output. In this section, sensitivity analysis is used as a tool in mathematical programming that allows one to visualize the effects of changed parameter values on the objective function. Sensitivity analysis is conducted on key parameters to clarify the operational efficiency of the proposed model and provide managerial perspectives. Considering that the GAMS programming software is one of the most efficient algorithms according to various criteria, it has been employed here for sensitivity analysis.

Thus, we create two scenarios for sensitivity analysis, investigating how the cost function changes in capacity and demand parameters for each section. The first scenario examines the effect of variations in the rate of recycled product production by waste poultry reprocessing factories on the objective function. In this case, the behavior of the objective function against the variation of parameters in Table 9 and Fig. 11 shows that an increase in the production rate of a recycled product in each section will lead to a reduction in the objective function value. On the other hand, there is a significant gap between the impact of reprocessing factories' n , a , and b on objective performance compared with other factors. As indicated, all factors considered positively correlate with an increase in the associated

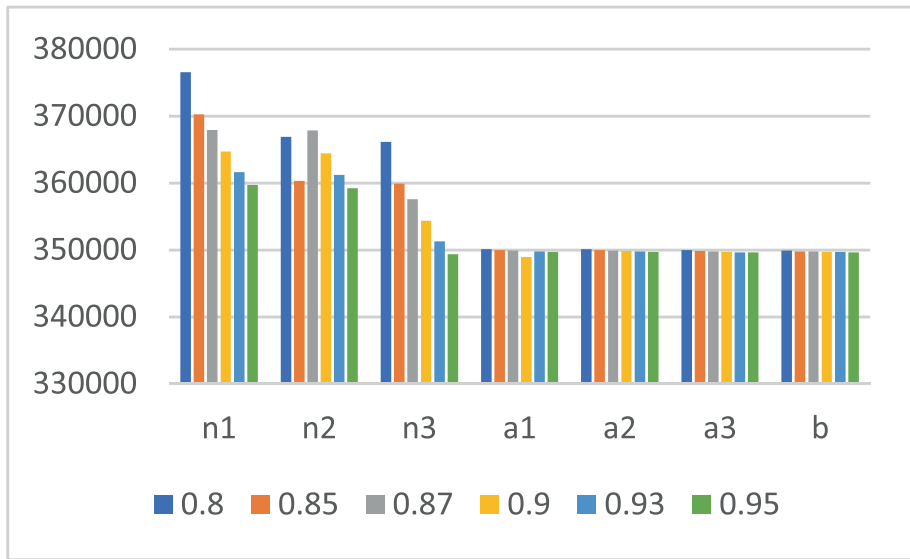


Fig. 11 Impact of changes in recycled product's production rates on the objective function

objective function value. Precise determination of the rate of recycled product production can prevent risky increases in the SC cost.

The second scenario examines how variations in customer demand for poultry and recycled products influence the supply chain cost. Demand values were varied between 5 and 25 tons, with all other parameters held constant. The results of the second scenario are presented in Table 10 and Fig. 12. These are individually calculated for customer demand and recycled product demand.

4.4 Computational results

Given that SC problems are NP-hard, the exact methods, like coding software GAMS, can solve it in small dimensions (only in 5 test problems), and for model validation, metaheuristic methods need to be employed. The validation of the presented algorithms is assessed using rigorous mathematical methods and GAMS programming. Table 11 shows the objective function results based on increasing model size and solving time, indicating that as the model size increases, the objective function also increases.

As previously mentioned, SC network issues in the real world are complex and thus NP-hard for large-scale cases (Tirkolaei et al., 2017). Employing precise methods to solve these problems, especially for larger-sized problems, will be time-consuming and inefficient. In this study, the GA, along with the SA and a hybrid of GA-SA, DE algorithm, and a hybrid of DE-SA are employed to find optimal solutions.

This stage aims to solve problems using the five proposed algorithms. The parameters remain constant while the size of the test problem changes during decomposition and analysis, according to Table 5. The performance evaluation of metaheuristic algorithms considers three criteria: Total Cost, Relative Percentage Deviation (RPD), and the computational time

Table 10 The impact of demand changes on total cost (Changes in Quantity)

Dp_{mp0}	5	342861.283858	Dp_{p1}	5	513204.274370
	10	348285.912140		10	693120.916940
	15	353853.647541		15	877001.990614
	20	359423.791971		20	1073006.338718
	25	365124.46227		25	1321727.538864
Dp_{p2}	5	504303.895712		5	513037.374833
	10	675076.198844		10	692899.638194
	15	849210.676438		15	876717.290949
	20	1031929.754279		20	1076953.305109
	25	1252524.439450		25	1339615.068621
	5	344996.682655		5	345015.724507
	10	352375.861936		10	352475.466538
	15	359755.291352		15	359980.977505
	20	367140.054096		20	367492.374267
	25	374526.926831		25	374982.245228
	5	345097.174249		5	356009.138128
	10	352554.431705		10	487943.102993
	15	360036.547694		15	655733.838589
	20	367515.403100		20	830686.533971
	25	375002.504990		25	967838.565789

of the algorithms. Tables 12, 13, and 14 show the behavior of an objective function for three criteria and different problem sizes. Different algorithms produce slightly different values.

This indicates different performances and optimizations of various algorithms for various problem sizes in the case of supply chain network optimization. For small-sized problems, the GA-SA hybrid provides the lowest total cost, as depicted by the consistently lower

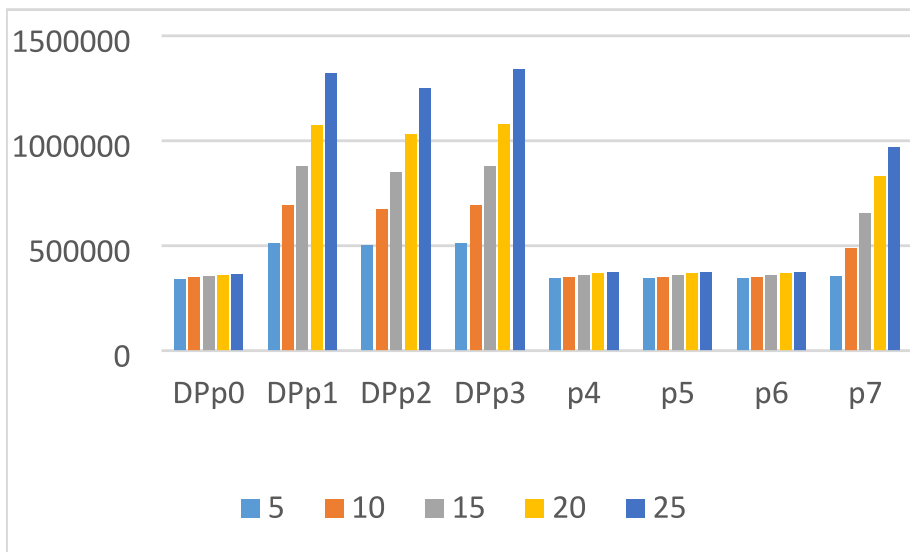
**Fig. 12** Impact of demand changes on the objective function

Table 11 Total cost for different problem sizes obtained by GAMS

Test problem number	Objective function values
1	348636
2	499597
3	677992
4	957780
5	1176827

value it achieves, hence good solution space exploration for less complex networks. DE has been proven to be the best method for medium-sized problems, with test problems from 9 through 16 resulting in superiority for networks of moderate complexity. In contrast, while handling large-sized problems - test problems 17 to 24, DE dominates, pointing toward the algorithm's robustness while dealing with the increase of scale and complexity in a more extensive supply chain network. Nevertheless, few exceptions exist, such as SA performing better than other algorithms over a few small-sized problems (test problem 8). These findings further support the view that applying the appropriate algorithm in the context of problem size and complexity is the key to optimum supply chain network design.

Overall, the DE algorithm can be superior to optimizing supply chain network designs for various problem sizes. For test problems 9–24, of medium- and large-sized problems, the DE algorithm has always obtained the minimum total cost, reflecting its robustness and efficiency in handling the complexities of a larger-sized network. In certain specific scenarios, it can be observed from Table 14 that other algorithms outperform the DE algo-

Table 12 Total cost for different problem sizes obtained by different algorithms

	SA	GA	GASA	DE	DESA
1	348637	348637	348637	348637	348637
2	499598	499598	499598	499598	499598
3	677993	677993	677993	677993	677993
4	957780	957780	957780	957780	1010277
5	1275629	1261748	1176827	1247386	1368243
6	1503742	1565657	1491469	1584817	1662219
7	2055057	2148384	2033458	2128235	2126054
8	2165581	2433160	2222245	2310112	2462586
9	2772919	2877203	2637650	2664706	3042359
10	3373289	3446493	3447478	3037605	3635695
11	3945821	4164434	4035000	3843458	4197074
12	4328755	4561927	4547989	4180012	4898092
13	4911931	5095789	4936603	4788559	5123604
14	5924425	6494606	5939927	5303480	6162882
15	7069680	7141821	7527411	6443937	7256854
16	8493074	8385978	8777918	7527207	8711381
17	11272063	10506786	11042695	9610474	11838952
18	13966286	13436422	13433792	11873050	13583518
19	15799492	15191210	16113395	13663136	16355569
20	18637244	17271862	17953744	15611532	18048615
21	20503047	18934675	20073649	17177156	20182403
22	22861301	21279841	22102581	18923020	22463715
23	24830934	23833136	24795320	20564800	24651552
24	27241757	26830780	26701430	22929427	26567951

Table 13 RPD of the objective function results by different algorithms

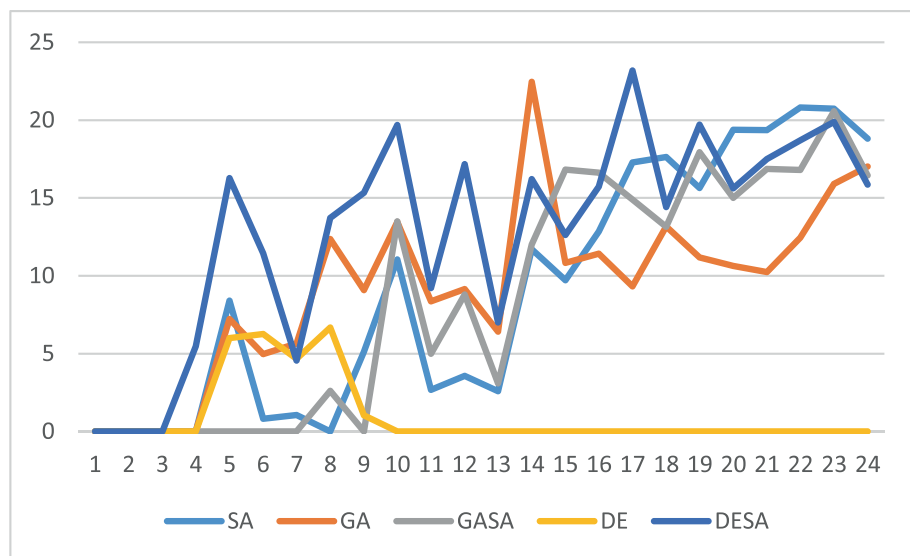
	SA	GA	GASA	DE	DESA
1	0	0	0	0	0
2	0	0	0	0	0
3	0	0	0	0	0
4	0	0	0	0	5.481066087
5	8.395560736	7.216069018	0	5.995699595	16.26535921
6	0.822907173	4.974176355	0	6.258801755	11.44843697
7	1.062201707	5.651750881	0	4.660887461	4.553613047
8	0	12.35595909	2.616568886	6.674003791	13.71478534
9	5.128394732	9.082076596	0	1.025782432	15.34355948
10	11.05094754	13.46084195	13.49327951	0	19.68950278
11	2.663298808	8.35122616	4.983602046	0	9.200472324
12	3.55842936	9.136675742	8.803241619	0	17.17887856
13	2.576382634	6.415909449	3.09160951	0	6.996782646
14	11.70825711	22.45933724	12.00055549	0	16.20449481
15	9.710570194	10.83008808	16.81385221	0	12.61522953
16	12.83167462	11.40888855	16.61586675	0	15.73191656
17	17.289346	9.326403518	14.90270857	0	23.18800881
18	17.63014233	13.16739484	13.14524535	0	14.40630022
19	15.63591366	11.18392025	17.93335838	0	19.70581777
20	19.3812621	10.63527728	15.00308404	0	15.61078332
21	19.36229654	10.23172496	16.86247451	0	17.49560326
22	20.81211217	12.45477803	16.80260615	0	18.71104531
23	20.744834	15.89286625	20.57165381	0	19.87255876
24	18.80696571	17.0146102	16.45049035	0	15.86836097

gorithm. For example, for small-sized problems, the hybrid algorithm GA-SA came out on top compared to DE in test problems 1 through 8. In these scenarios, though, DE still tends to be very competitive and does not fall significantly behind the best-performing algorithm in the small-sized scenario, further establishing its versatility and reliability on various problem sizes. Figure 13. Compares the algorithms based on RPD. The last criterion used to assess algorithm performance is computational time. Figure 14 displays information regarding computational time for all algorithms, where it is notable that SA has the least computational time.

Table 14 presents the time different algorithms take to solve various-sized optimization problems. As the problem size increases from 1 to 24, the time taken by the algorithms has increased as the problem size increases due to the immense computational complexity of the problem. SA consistently shows a relatively shorter solution time than other algorithms for most problem sizes. However, DE shows the best solution times, especially for small problem sizes (test problems 2 to 7), and DESA reveals a longer time for most problems. This suggests that while SA may offer faster convergence to solutions, DESA and DE may require more computational resources to solve large-scale optimization problems. The overall solution time for all algorithms significantly increases from medium-sized problems to large ones, which reflects the increasing computational burdens while dealing with larger problem instances. It is clear from the analysis that there is a tradeoff between performance on cost optimization and time of computation, where DE is more efficient on cost, but it may require more computational resources.

Table 14 Obtained time by different algorithms

	SA	GA	GASA	DE	DESA
1	0.93897012	1.3498496	1.45371067	0.89895415	1.48195038
2	2.87875304	3.03355698	3.4943957	1.90671328	3.52489338
3	2.26767992	2.53634402	3.20944649	1.67308267	2.91778681
4	2.63041741	3.09761781	3.22803804	1.84976861	3.04313543
5	2.74688736	3.0442671	3.36949634	2.19523119	3.42691217
6	2.71022793	3.3900524	3.67161893	2.46459055	3.95638364
7	2.99315332	3.89175148	4.234839	2.98319149	4.91975694
8	3.24385843	4.29173323	4.57996348	3.52646146	5.51040902
9	2.66796082	4.10469362	4.56048069	5.58393192	7.50171887
10	2.92798354	4.32354354	5.01360331	6.51707867	9.15321184
11	3.50587995	5.13685029	5.92642428	8.36332756	11.30832515
12	4.02121288	5.94835614	6.90466091	10.51835826	14.35372367
13	4.54824563	6.75851451	7.86032274	12.6567131	18.04468709
14	5.07937108	7.80376561	9.00013578	15.73097341	22.2262338
15	5.99517754	9.21384137	10.72522191	18.7416029	26.4655423
16	7.25998542	11.25291902	12.82289694	25.85614125	35.36275334
17	4.9291594	16.73570804	17.17027028	30.35238133	68.20019511
18	6.03202124	19.78864454	21.01219963	39.96933317	93.57602018
19	7.12337885	23.33320973	26.078744	51.18780255	121.4678554
20	8.26544841	27.81434159	30.6037851	63.44639379	149.6966661
21	10.10571488	33.51804181	38.14983022	84.44078268	199.8445123
22	11.92944335	39.56919887	45.9097988	106.5431364	254.7648508
23	13.95669656	47.04671965	54.52756173	128.2193717	300.76733
24	17.26663762	59.83810179	69.63713976	155.7965688	374.0900096

**Fig. 13** Algorithms results in terms of RPD

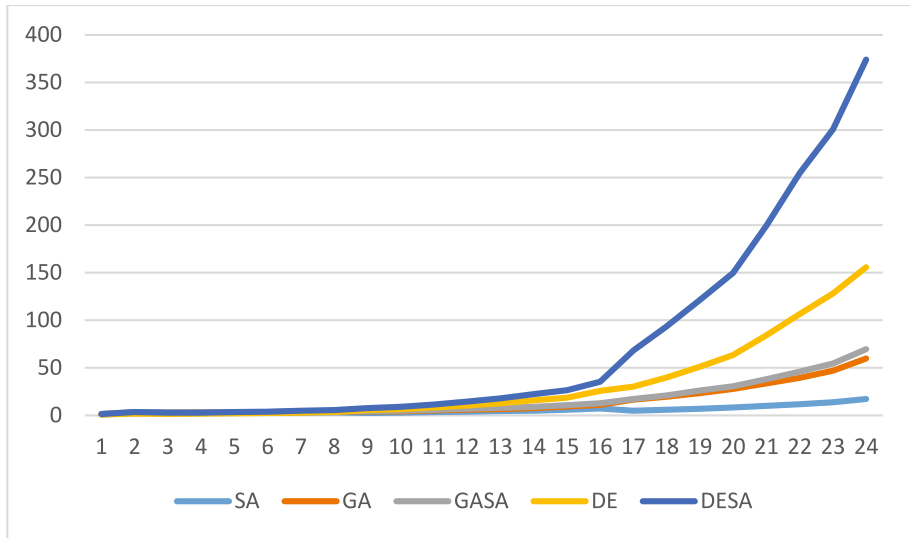


Fig. 14 Algorithms solution time

4.5 Managerial insight

The primary goal of this research was to introduce an SC for poultry products originating from poultry farming centers. The uniqueness of this model stems from its innovative method of recycling poultry waste generated by these centers. Slaughterhouses, poultry processing factories, and distribution centers are raw material sources for poultry waste recycling plants. Implementing the developed SC network could help officials and food industry managers make strategic decisions by providing guidance. Hence, the findings of this model can aid investors and officials in determining the best locations for optimal distribution and transfer of poultry products throughout the network. Besides, the poultry industry may enhance its business activity by adding poultry waste recycling and offering even higher environmental bonuses if such technology is not widely provided.

Any other product could be substituted for poultry, and the same problem could be related to any other industry. The new supply chain model proposed in this study, in which recycling of poultry waste is of primary importance and where distribution networks are optimized, could theoretically provide a versatile framework for other industries dealing with similar kinds of products. Adapting this model to other food industries, like dairy, seafood, and fresh produce, among others, will empower the stakeholders to apply similar approaches to enhance sustainability and operational efficiency.

Another managerial concept obtained from this study relates to poultry processing plants. The plant managers can use this model to improve operational and fixed transport costs and manage the production and flow of SC operations. An optimized SC, therefore, has a minimum cost and offers a better service level, comprising more value than a comparable SC. For example, managers can obtain coordination between their production market demand and capacity with the constraints of the model. On the other hand, sensitivity analysis of the parameters involved in the proposed model can also provide some exciting insights for SC

managers. SC decision-makers always seek effective methods to address issues and make successful decisions. This study introduced some metaheuristic algorithms that poultry and bird production industry managers can employ to address specific or similar network design issues.

5 Conclusion

In the current competitive situation, a lean SC for poultry products at low costs is the primary requirement, more so for poultry meat with a capacity for significant exports. However, poultry by-products threaten environmental health; they could be recycled into the value chain using processes like manure production, composting, and oil extraction. This paper focused on designing a closed-loop poultry supply chain, intending to reduce costs and mitigate the negative impacts of by-products. A mixed-integer programming model was formulated and verified with precise mathematical methods. In addition, three metaheuristic algorithms were used to solve the model: GA, SA, and DE, using two-hybrid approaches, GA-SA and DE-SA. The intensities of these algorithms were carefully tuned using the Taguchi experimental design method. The results indicate that the GA-SA algorithm performs better when the problems are small, whereas DE performs better in medium and larger problems.

Furthermore, the developed model provides a flexible framework for CLSC construction across several industries, driving sustainability and operational efficiency. Provided that the stakeholders adapt the model to other sectors, such as red meat, seafloor, or companies dealing in fresh produce, they would attain similar results in terms of sustainability. That is why this study has become an innovative one: it has the potential to help cut huge costs and environmental impacts through proper management of by-products, which offers considerable benefits to the poultry industry and may go far.

5.1 Future research

Future studies will enhance this model by including multi-objective functions that consider issues about the quality of the product, like shelf life and delivery time. Multi-period settings of the model and dimensions of sustainability—social, environmental, and economic—make it much more thorough. Another important thing is to handle uncertainty due to real-world scenarios; future models can be developed using robust optimization methods or stochastic control to handle the unpredictability of market demands. Further research into human factors—such as psychological solution approaches—could also be helpful.

Data availability The data supporting this study's findings are available on request from the first author.

References

- Amorim, P., Meyr, H., Almeder, C., & Almada-Lobo, B. (2013). Managing perishability in production-distribution planning: A discussion and review. *Flexible Services and Manufacturing Journal*, 25(3), 389–413.

- Baghalian, A., Rezapour, S., & Farahani, R. Z. (2013). Robust supply chain network design with service level against disruptions and demand uncertainties: A real-life case. *European Journal of Operational Research*, 227(1), 199–215.
- Baghizadeh, K., Pahl, J., & Hu, G. (2021). Closed-loop supply chain design with sustainability aspects and network resilience under uncertainty: Modelling and application. *Mathematical Problems in Engineering*, 2021, 1–23.
- Borodin, V., Bourtembourg, J., Hnaien, F., & Labadie, N. (2016). Handling uncertainty in agricultural supply chain management: A state of the art. *European Journal of Operational Research*, 254(2), 348–359.
- Çevik, A., Weber, G. W., Eyüboğlu, B. M., Oğuz, K. K., Alzheimer's Disease Neuroimaging Initiative. (2017). Voxel-MARS: A method for early detection of Alzheimer's disease by classification of structural brain MRI. *Annals of Operations Research*, 258, 31–57.
- Cheraghalipour, A., Paydar, M. M., & Hajiaghaei-Keshteli, M. (2018). A bi-objective optimization for citrus closed-loop supply chain using Pareto-based algorithms. *Applied Soft Computing*, 69, 33–59.
- Conway, A. (2015). Global egg consumption to rise worldwide through 2024. Wattagnet. com. Publicada el, 23. <https://www.wattagnet.com/broilers-turkeys/article/15515897/global-egg-consumption-to-rise-worldwide-through-2024-wattagnet>
- De Vries, W., Kros, J., Kroeze, C., & Seitzinger, S. P. (2013). Assessing planetary and regional nitrogen boundaries related to food security and adverse environmental impacts. *Current Opinion in Environmental Sustainability*, 5(3–4), 392–402.
- Fallahi, A., Mokhtari, H., & Niaki, S. T. A. (2021). Designing a closed-loop blood supply chain network considering transportation flow and quality aspects. *Sustainable Operations and Computers*, 2, 170–189.
- Fallahi, A., Mousavian Anaraki, S. A., Mokhtari, H., & Niaki, S. T. A. (2022). Blood plasma supply chain planning to respond COVID-19 pandemic: A case study. *Environment, Development and Sustainability*. <https://doi.org/10.1007/s10668-022-02793-7>
- FAO. (2020). Emissions due to agriculture: Global, regional and country trends. <https://fao.org>
- Fathollahi-Fard, A. M., Hajiaghaei-Keshteli, M., & Mirjalili, S. (2020). A set of efficient heuristics for a home healthcare problem. *Neural Computing and Applications*, 32, 6185–6205.
- Gholipour, A., Sadegheih, A., Mostafaeipour, A., & Fakhrazad, M. B. (2023). Designing an optimal multi-objectivemodel for a sustainable closed-loop supply chain: A case study of pomegranate in Iran. *Environment, Development and Sustainability*. <https://doi.org/10.1007/s10668-022-02868-5>
- Goodarzian, F., Ghasemi, P., Gonzalez, E. D. S., & Tirkolaei, E. B. (2023). A sustainable-circular citrus closed-loop supply chain configuration: Pareto-based algorithms. *Journal of Environmental Management*, 328, 116892.
- Govindan, K., & Soleimani, H. (2017). A review of reverse logistics and closed-loop supply chains: A journal of cleaner production focus. *Journal of Cleaner Production*, 142, 371–384.
- Govindan, K., Jafarian, A., & Nourbakhsh, V. (2015). Bi-objective integrating sustainable order allocation and sustainable supply chain network strategic design with stochastic demand using a novel robust hybrid multi-objective metaheuristic. *Computers & Operations Research*, 62, 112–130.
- Haghbayan, M., Karimi, B., Mozdgir, A., & Abbaspanah, B. (2023). Increasing the efficacy of umbilical cord blood banking using machine learning algorithms: A case study from Royan cord blood bank. *Scientia Iranica*. <https://doi.org/10.24200/sci.2023.61068.7132>
- Hamdan, B., & Diabat, A. (2020). Robust design of blood supply chains under risk of disruptions using Lagrangian relaxation. *Transportation Research Part E: Logistics and Transportation Review*, 134, Article 101764.
- Hasani, A., Zegordi, S. H., & Nikbakhsh, E. (2012). Robust closed-loop supply chain network design for perishable goods in agile manufacturing under uncertainty. *International Journal of Production Research*, 50(16), 4649–4669.
- Holland, J. H. (1992). Genetic algorithms. *Scientific American*, 267(1), 66–73.
- Hosseini Baboli, S. A., Arabkoohsar, A., & Seyed, I. (2023). Numerical modeling and optimization of pressure drop and heat transfer rate in a polymer fuel cell parallel cooling channel. *Journal of the Brazilian Society of Mechanical Sciences and Engineering*, 45(4), Article 201.
- Jouzdani, J., & Govindan, K. (2021). On the sustainable perishable food supply chain network design: A dairy products case to achieve sustainable development goals. *Journal of Cleaner Production*, 278, Article 123060.
- Karadağ, İ., Keskin, M. E., & Yiğit, V. (2021). Re-design of a blood supply chain organization with mobile units. *Soft Computing*, 25(8), 6311–6327.
- Karimi, A., Mozdgir, A., Tashakkori, R., Bagheri Tofighi, A., & Bozorgzadehvostakolaei, S. (2024). Predicting antenatal depression during 3rd trimester: A machine learning approach with feature selection and TOPSIS ranking. *Scientia Iranica*. <https://doi.org/10.24200/sci.2024.63194.8288>
- Kaya, O., & Ozkok, D. (2020). A blood bank network design problem with integrated facility location, inventory and routing decisions. *Networks and Spatial Economics*, 20(3), 757–783.

- Keshavarz-Ghorbani, F., & Pasandideh, S. H. R. (2023). Designing a multi-objective closed-loop supply chain for multi-period multi-generational products with social impacts considerations. *Computers & Industrial Engineering*, 177, Article 109056.
- Khalilpourazari, S., & Arshadi Khamseh, A. (2019). Bi-objective emergency blood supply chain network design in earthquake considering earthquake magnitude: A comprehensive study with real world application. *Annals of Operations Research*, 283(1), 355–393.
- Khalili-Fard, A., Parsaee, S., Bakhshi, A., Yazdani, M., Aghsami, A., & Rabbani, M. (2024). Multi-objective optimization of closed-loop supply chains to achieve sustainable development goals in uncertain environments. *Engineering Applications of Artificial Intelligence*, 133, 108052.
- Khalilpourazari, S., Soltanzadeh, S., Weber, G. W., & Roy, S. K. (2020). Designing an efficient blood supply chain network in crisis: Neural learning, optimization and case study. *Annals of Operations Research*, 289, 123–152.
- Kirkpatrick, S., Gelatt, C. D., Jr., & Vecchi, M. P. (1983). Optimization by simulated annealing. *Science*, 220(4598), 671–680.
- Kuter, S., Akyurek, Z., & Weber, G. W. (2018). Retrieval of fractional snow covered area from MODIS data by multivariate adaptive regression splines. *Remote Sensing of Environment*, 205, 236–252.
- Li, Y., Arulnathan, V., Heidari, M. D., & Pelletier, N. (2022). Design considerations for net zero energy buildings for intensive, confined poultry production: A review of current insights, knowledge gaps, and future directions. *Renewable and Sustainable Energy Reviews*, 154, Article 111874.
- Mavi, R. K., Shekarabi, S. A. H., Mavi, N. K., Arisian, S., & Moghdani, R. (2023). Multi-objective optimisation of sustainable closed-loop supply chain networks in the tire industry. *Engineering Applications of Artificial Intelligence*, 126, 107116.
- Mirzaei, M. G., Gholami, S., Rahmani, D., Goodarzi, F., & Khanchehazarrin, S. (2024). Robust optimization in an agricultural closed-loop supply chain network design with a price and freshness-dependent demand: hybrid rat with particle swarm optimization algorithm. *Environment, Development and Sustainability*, 1–36. <https://doi.org/10.1007/s10668-024-05296-9>
- Mokhtari, H., & Mozdgir, A. (2015). A bi-objective approach to an assembly line re-balancing problem: Model and differential evolution algorithms. *International Journal of Industrial Engineering & Production Research*, 26(2), 105–117.
- Mosallanezhad, B., Hajiaghaci-Keshteli, M., & Triki, C. (2021). Shrimp closed-loop supply chain network design. *Soft Computing*, 25, 7399–7422.
- Mozdgir, A., Mahdavi, I., Badeleh, I. S., & Solimanpur, M. (2013). Using the Taguchi method to optimize the differential evolution algorithm parameters for minimizing the workload smoothness index in simple assembly line balancing. *Mathematical and Computer Modelling*, 57(1–2), 137–151.
- Navaei, J., Mozdgir, A., & Hidaji, H. (2010). Two-stage assembly flow-shop scheduling problem with bi-objective of number of tardy and makespan minimization. In ICAI 2010: proceedings of the 2010 international conference on artificial intelligence (Las Vegas NV, July 12–15, 2010) (pp. 704–709). <http://pascal-francis.inist.fr/vibad/index.php?action=getRecordDetail&idt=26080642>
- Okay, N.C., Sencer, A., & Taskin, N. (2024). Quantitative indicators for environmental and social sustainability performance assessment of the supply chain. *Environment, Development and Sustainability*. <http://doi.org/10.1007/s10668-024-05210-3>
- Ostojic, P. (2016). Pumpsand circular economy. *World Pumps*, 2016(1), 30–33.
- Özmen, A., Kropat, E., & Weber, G. W. (2014). Spline regression models for complex multi-modal regulatory networks. *Optimization Methods and Software*, 29(3), 515–534.
- Özmen, A., Kropat, E., & Weber, G. W. (2017). Robust optimization in spline regression models for multi-model regulatory networks under polyhedral uncertainty. *Optimization*, 66(12), 2135–2155.
- Pishvae, M. S., Rabbani, M., & Torabi, S. A. (2011). A robust optimization approach to closed-loop supply chain network design under uncertainty. *Applied Mathematical Modelling*, 35(2), 637–649.
- Rajabi-Kafshgar, A., Gholian-Jouybari, F., Seyedi, I., & Hajiaghaci-Keshteli, M. (2023). Utilizing hybrid metaheuristic approach to design an agricultural closed-loop supply chain network. *Expert Systems with Applications*, 217, Article 119504.
- Rajabi-Kafshgar, A., Seyedi, I., & Tirkolaee, E. B. (2025). Circular closed-loop supply chain network design considering 3D printing and PET bottle waste. *Environment, Development and Sustainability*. <https://doi.org/10.1007/s10668-024-04767-3>
- Rohmer, S. U. K., Gerdessen, J. C., & Claassen, G. D. H. (2019). Sustainable supply chain design in the food system with dietary considerations: A multi-objective analysis. *European Journal of Operational Research*, 273(3), 1149–1164.
- Salehi-Amiri, A., Zahedi, A., Gholian-Jouybari, F., Calvo, E. Z. R., & Hajiaghaci-Keshteli, M. (2022). Designing a closed-loop supply chain network considering social factors; A case study on avocado industry. *Applied Mathematical Modelling*, 101, 600–631.

- Sel, C., Bilgen, B., Bloemhof-Ruwaard, J. M., & van der Vorst, J. G. (2015). Multi-bucket optimization for integrated planning and scheduling in the perishable dairy supply chain. *Computers & Chemical Engineering*, 77, 59–73.
- Seyedi, I., Maleki-Daronkolaei, A., & Kalashi, F. (2012). Tabu search and simulated annealing for new three-stage assembly flow shop scheduling with blocking. *Interdisciplinary Journal of Contemporary Research in Usiness*, 4(8), 394–402.
- Seyedi, I., Mirzazadeh, S., Maleki-Daronkolaei, A., Mukhtar, M., & Sahran, S. (2016). An inventory model with reworking and setup time to consider effect of inflation and time value of money. *Journal of Engineering Science and Technology*, 11(3), 416–430.
- Seyedi, I., Hamed, M., & Tavakkoli-Moghaddam, R. (2022). Optimization for a truck scheduling problem in multi-door cross-docking with learning effect and deteriorating jobs. *Journal of Transportation Research*, 19(71), 183–206. <https://doi.org/10.22034/tri.2022.119317>
- Seyedi, I., & Maleki-Daronkolaei, A. (2013). Solving a two-stage assembly flowshop scheduling problem to minimize the mean tardiness and earliness penalties by three meta-heuristics. *Caspian Journal of Applied Sciences Research*, 2(4). https://www.researchgate.net/profile/Aref-Maleki/publication/272660858_Solving_a_two-stage_assembly_flowshop_scheduling_problem_to_minimize_the_mean_tardiness_and_earliness_penalties_by_three_meta-heuristics/links/54eaf4a10cf27a6de11548fe/Solving-a-two-stage-assembly-flowshop-scheduling-problem-to-minimize-the-mean-tardiness-and-earliness-penalties-by-three-meta-heuristics.pdf
- Seyfi-Shishavan, S. A., Donyatalab, Y., Farrokhzadeh, E., & Satoglu, S. I. (2021). A fuzzy optimization model for designing an efficient blood supply chain network under uncertainty and disruption. *Annals of Operations Research*, (1), 55. <https://doi.org/10.1007/s10479-021-04123-y>
- Shekarabi, H., Vali-Siar, M. M., & Mozdgir, A. (2024). Food supply chain network design under uncertainty and pandemic disruption. *Operational Research*, 24(2), 26.
- Shekarian, E. (2020). A review of factors affecting closed-loop supply chain models. *Journal of Cleaner Production*, 253, Article 119823.
- Shokouhifar, M., Naderi, R., Goli, A., Gultom, P., Nikabadi, M. S., & Weber, G. W. (2024). Metaheuristic-driven extended exergy accounting for sustainable closed-loop food supply chain management. *Computers & Industrial Engineering*, 191, Article 110148.
- Taguchi, G. (1986). Introduction to quality engineering: designing quality into products and processes (No. 658.562 T3). <https://trid.trb.org/View/1179550>
- Tashakkori, R., Mozdgir, A., Karimi, A., & BozorgzadehVostaKolaei, S. (2024). The prediction of NICU admission and identifying influential factors in four different categories leveraging machine learning approaches. *Biomedical Signal Processing and Control*, 90, Article 105844.
- Tavana, M., Kian, H., Nasr, A. K., Govindan, K., & Mina, H. (2022). A comprehensive framework for sustainable closed-loop supply chain network design. *Journal of Cleaner Production*, 332, Article 129777.
- Weber, G. W., Taylan, P., Alparslan-Gök, S. Z., Özögür-Akyüz, S., & Akteke-Öztürk, B. (2008). Optimization of gene-environment networks in the presence of errors and uncertainty with Chebyshev approximation. *TOP*, 16, 284–318.
- Yang, Y., & Chen, Y. (2021). Designing closed-loop supply chain network for high-salvage perishable products under uncertain environment. *IEEE Access*, 9, 102236–102250.
- Yavari, M., & Geraeli, M. (2019). Heuristic method for robust optimization model for green closed-loop supply chain network design of perishable goods. *Journal of Cleaner Production*, 226, 282–305.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

Authors and Affiliations

Zahra Akbari-Aghghaleh¹ · Ashkan Mozdgir¹  · Iman Seyedi² · Enza Messina²



Ashkan Mozdgir

A.Mozdgir@khu.ac.ir

Zahra Akbari-Aghghaleh

Zahra.Akbari1995@yahoo.com

Iman Seyedi

SeyedIman.Seyedi@unimib.it

Enza Messina

enza.messina@unimib.it

¹ Department of Industrial Engineering, Faculty of Engineering, Kharazmi University, Tehran, Iran

² Department of Informatics, Systems and Communication, University of Milano-Bicocca, 20125 Milan, Italy